

## Review article

# Evolution, topics and relevant research methodologies in business intelligence and data analysis in the academic management of higher education institutions. A literature review

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## ARTICLE INFO

## Keywords:

Academic analytics  
Bibliometric analysis  
Bibliometrix  
Business intelligence  
Data analytics  
Educational data mining  
Higher education  
Learning analytics

## ABSTRACT

In the digital era, Business Intelligence (BI) and data analytics have become essential for optimizing academic management in higher education institutions. This bibliometric study analyzed 755 Scopus-indexed publications (2019–2023) using RStudio, Biblioshiny, and Microsoft Excel to elucidate key themes, influential authors, and emerging research trends. Learning analytics, educational data mining, and BI applications such as dropout prediction systems, tailored distance education strategies, and machine learning models for institutional performance predominate in the field. High-impact journals, including the British Journal of Educational Technology and the Journal of Learning Analytics, play crucial roles with contributions from scholars such as Christothea Herodotou and Bart Rienties. Thematic analysis revealed ten clusters emphasizing predictive modeling, educational innovation, and online learning. Geographic trends highlight the predominance of research in the United States and Europe, underscoring the necessity for greater inclusivity in underrepresented regions such as Africa and South America. While quantitative methodologies prevail, this study emphasizes the significance of qualitative approaches to capture nuanced impacts and ethical considerations, including privacy, equity, and bias mitigation. Future research must adopt interdisciplinary methodologies to address systemic challenges, foster context-sensitive, equitable BI solutions that drive innovation, and enhance decision making across diverse educational environments.

## 1. Introduction

In today's digital context, academic management in higher education institutions is undergoing a significant transformation, driven by advances in business intelligence (BI) and data analytics. These disciplines have revolutionized informed decision-making within academia, becoming fundamental to the improvement and understanding of processes ranging from strategic planning to operational decisions in higher education institutions.

BI and data analytics, which are intrinsically linked fields, focus on the use of data to facilitate strategic and operational decision-making. BI is specifically dedicated to the collection, analysis, and presentation of data to support strategic decisions within an organization [1], employing techniques such as data mining [2], visualization [3], and reporting. Data analytics, on the other hand, employs statistical and quantitative methods to identify patterns and extract meaningful insights [4], encompassing techniques from descriptive [5,6] to predictive [7,8] and

prescriptive [9], to discover trends, make predictions, and optimize business processes. Together, both fields are crucial for achieving competitive advantage, improving efficiency, and fostering growth [4, 10].

Additionally, academic analytics represents the application of these analytical techniques in education, aimed at optimizing both academic performance and learning processes. Through educational data mining (EDM) and learning analytics, large volumes of educational data have been explored to solve critical questions in education [2]. This approach allows not only understanding and improving academic performance but also optimizing the learning experience by analyzing feedback, assessments, and student outcomes [11,12].

Educational data mining (EDM) is an interdisciplinary field that uses techniques such as machine learning, data analysis, and visualization to extract valuable information from educational data such as students' interactions with online learning systems and their academic performance [13,14]. Several studies have explored the use of EDM for various

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<https://doi.org/10.1016/j.rineng.2024.103782>

Received 8 October 2024; Received in revised form 16 December 2024; Accepted 18 December 2024

Available online 19 December 2024

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purposes, applying techniques such as logistic regression [3,15], clustering [16], support vector machines [13,17], decision trees [13,18,19], random forests [14,20], and artificial neural networks [15] to predict student performance, recommend resources, evaluate learning materials, and model student behavior. The primary objective of these fields is to enhance educational results by implementing data-driven insights and personalized learning techniques [9,17,21]. Furthermore, studies such as 'Insights from the COVID-19 Pandemic' illustrate how data mining techniques were instrumental in managing educational disruptions, providing insights into scalable solutions for crises [22].

In the context of Higher Education Institutions, university administrators have increasingly relied on BI and data analytics to facilitate strategic decision-making [10,23,24]. These resources have become essential for addressing key challenges such as optimizing resources [25], improving student retention [26]–[27], adapting academic programs to labor market dynamics and student needs, and assessing performance [10] and institutional efficiency [28]. Effective implementation of data analytics provides crucial insights for analysis and improvement in critical areas, such as optimizing educational processes [28], administrative [29], enriching the student experience [27, 30], fiscal management, and improving learning environments [31,32], thereby strengthening decision-making in academic planning and resource management.

Although current research in BI and data analytics in Higher Education Institutions (HEIs) has shed light on applications for improving institutional performance and management, there is a vast field yet to be explored in areas such as academic management [33], admissions [15], human resources [33], finance [34], budgeting, research [35], administrative operations, and institutional development [36,37]. Identifying trends and challenges in these less explored domains opens avenues for future research that could provide significant advances in the use of analytics for higher education optimization. This approach not only broadens the spectrum of application of BI and data analytics in HEIs but also promotes institutional efficiency, management, and performance through the strategic use of data.

Several studies have investigated BI and data analyses in higher education. However, it is crucial to conduct an updated bibliometric review in order to gain a more comprehensive understanding of its global progression. It is crucial to identify major themes and trends, focusing on widely cited works as well as examining the geographical distribution of publications and their connection to the educational sector. Future research should prioritize the identification of methodological approaches and dominant lines of research. This analysis facilitates a comprehensive understanding of their impact on academic management. The importance lies in providing a holistic perspective on the transformation of higher education and establishing a comprehensive conceptual framework for understanding these views.

### 1.1. Evolution of business intelligence (BI) tools in education

The progression of Business Intelligence (BI) and data analysis tools in the academic management of higher education institutions has significantly transformed strategic decision-making processes. These advanced technologies facilitate the management of extensive data volumes to personalize learning experiences, optimize resource allocation, and enhance institutional efficacy. This section examines the critical developmental stages of BI in academic administration by tracing its evolution from initial descriptive applications to its current integration in predictive and ethical frameworks.

The development of BI and data analytics tools in education draws heavily on foundational research in predictive analytics and data mining. Early studies provided essential frameworks that remain relevant to current applications. For instance, Mengash (2020) highlighted the application of data mining techniques to predict student performance and optimize decision-making processes in university admissions systems, illustrating how early innovations laid the groundwork for more

efficient, data-driven institutional strategies [38]. Similarly, Adekitan and Salau (2019) demonstrated the impact of early performance indicators on final graduation outcomes, showcasing how BI tools have scaled to address complex challenges such as retention and academic performance optimization [39].

Further advancements in BI tools during the mid-2010s incorporated machine learning algorithms to enhance predictive models and support personalized educational approaches. These techniques not only improved institutional decision-making but also contributed to refining adaptive learning systems. Complementing these predictive methods, Rajabalee and Santally (2021) emphasized the integration of mixed methodologies to provide a holistic view of learner satisfaction and engagement, further bridging qualitative and quantitative approaches in BI research [40].

This evolution of BI tools also reflects growing attention to governance and ethical considerations. From a governance perspective, Combata Niño et al. (2020) proposed frameworks to ensure equitable and transparent implementation of BI systems in higher education institutions [37]. These frameworks address critical concerns such as access disparities, scalability, and data privacy, demonstrating how foundational research has informed the alignment of advanced machine learning techniques with institutional strategies.

The descriptive phase (2010–2015) saw BI tools primarily focused on data visualization and descriptive analytics, utilized to monitor essential metrics, such as attendance and academic performance. These pioneering tools have established a foundation for sophisticated applications. Akçapinar et al. (2019) noted that the implementation of early warning systems based on descriptive analytics enabled the efficient identification of at-risk students, marking a significant advancement in educational management [41].

Building on this foundation, the period from 2016 to 2019 witnessed BI tools evolving to incorporate predictive modeling and machine learning. These technological advancements have enabled the forecasting of academic outcomes and customization of educational approaches, addressing challenges such as student retention. Perez-Colado et al. (2019) investigated the streamlined development of educational games as an innovative tool merging gamification elements with BI to enhance student engagement and learning outcomes [42].

Since 2020, BI implementation has embraced ethical and social frameworks that balance intensive data utilization with principles of privacy and equity. Shukor and Abdullah (2019) emphasized how learning analytics can be designed to enhance instruction in massive online learning environments (MOOCs) while safeguarding student rights and needs [43]. Suleman and Chigeza (2019) highlighted the role of these tools in fostering increased participation among first-year higher education students [44]. The introduction of more learner-centric technologies, such as the analytical models of reading behavior outlined by Boticki et al. (2019), has revolutionized education by providing insights into individual engagement patterns and learning preferences [45].

The continued advancement of BI in academic administration has not only refined strategic decision-making, but also generated key areas of innovation, including gamification and bibliometric analysis. Nevertheless, issues such as data privacy and platform compatibility underscore the need for sustainable approaches to maintaining the inclusivity and effectiveness of these technologies. Table 1 examines the most pertinent issues and methodologies associated with BI and data analytics in higher education academic management, offering a framework for understanding how these technologies can continue to enhance educational administration in an increasingly digitized landscape.

This article aims to present a bibliometric review of studies published in Scopus from 2019 to December 2023 to identify the main study variables [62], including publications, keywords, citations, authors, and institutions, to explore the interaction of variables in academic management in higher education [63,64]. In addition, it seeks to identify gaps, knowledge gaps, critical points, opportunities, and limitations for

**Table 1**

Evolution of Business Intelligence (BI) tools in education (2019–2023).

| Year | Historical Context                                                                                                     | Focus Areas                                                                               | Key Innovations                                                              | Key Challenges                                                                      | Most Relevant Studies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | BI Tools Applied                                                    |
|------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 2019 | Initial introduction of BI tools for basic data analysis and visualization. Limited access and institutional adoption. | Introduction of foundational BI tools for visualization and descriptive analytics.        | Simplified game-based learning tools.                                        | Limited access to advanced BI tools and low technological integration.              | 1. Akçapınar et al. (2019) - Using learning analytics to develop early-warning system for at-risk students [41];<br>2. Pérez-Colado et al. (2019) - Simplifying the creation of adventure serious games [42];<br>3. Suleman and Chigeza (2019) - Enhancing participation of first-year education students [44];<br>4. Shukor, N.A. and Abdullah, Z. (2019). Using learning analytics to improve MOOC instructional design [43].<br>5. Boticki, I. et.al. (2019). E-book user modelling through learning analytics: The case of learner engagement and reading styles [45].                                                                                                                                                               | Data visualization, Learning analytics, Simplified game tools.      |
| 2020 | Expansion of predictive analytics to identify at-risk students and improve learning outcomes.                          | Adoption of governance frameworks and predictive analytics for institutional improvement. | Machine learning algorithms and data mining for early interventions.         | Resistance to change and lack of trained personnel in advanced BI methods.          | 1. Combita et al. (2020) - Business intelligence governance framework in a university [37];<br>2. Hooshyar et al. (2020) - Mining educational data to predict students' performance [46];<br>3. Mengash (2020) - Using data mining techniques to predict student performance to support decision making in university admission systems [38];<br>4. Aydoğdu, Ş. (2020). Predicting student final performance using artificial neural networks in online learning environments [47].<br>5. Yang, Y. et.al. (2020). Predicting course achievement of university students based on their procrastination behaviour on Moodle [17].                                                                                                          | Predictive analytics, Machine learning, Data mining.                |
| 2021 | Integration of BI into digital transformation strategies with a focus on governance and ethics.                        | Exploration of social and ethical dimensions of BI in education.                          | Addressing social and ethical impacts of BI in education.                    | Balancing data collection with student privacy and addressing regional disparities. | 1. García-Peñalvo F.J. (2021) - Avoiding the dark side of digital transformation in teaching. An institutional reference framework for e-learning in higher education [48];<br>2. Larabi-Marie-Sainte, S. et al. (2021). The impact of timetable on student's absences and performance [49].<br>3. Kokoç, M. et.al (2021). Unfolding students' online assignment submission behavioral patterns using temporal learning analytics [50].<br>4. Jamjoom, M.M. et al. (2021). Early prediction for at-risk students in an introductory programming course based on student self-efficacy [51].<br>5. Pazmiño-Maji, R. et al. (2021). Learning analytics in Ecuador: A systematic review supported by statistical implicative analysis [52]. | Theoretical frameworks, Ethical analytics, Educational data mining. |
| 2022 | Consolidation of BI use with dashboards and real-time monitoring tools for academic performance.                       | Development of predictive and visual analytics to optimize academic outcomes.             | Combination of qualitative and quantitative research methods for evaluation. | Scalability and interoperability challenges with large student cohorts.             | 1. Polat, E. et.al. (2022). Examining the effect of feedback type and gender on computing achievements, engagement, flipped learning readiness, and autonomous learning in online flipped classroom [53].<br>2. Du, G. et al. (2022). Construction and analysis of school moral education system based on big data technology [54].<br>3. Deho, O.B. et al. (2022). How do the existing fairness metrics and unfairness mitigation algorithms contribute to ethical learning analytics? [55].<br>4. Liu, C. et.al. (2022). A predictive model for student achievement using spiking neural networks based on                                                                                                                             | Visual analytics, Dashboards, Predictive tools.                     |

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Table 1 (continued)

| Year | Historical Context                                                                  | Focus Areas                                                        | Key Innovations                                             | Key Challenges                                                              | Most Relevant Studies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | BI Tools Applied                                                     |
|------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 2023 | Global adoption and systematic evaluation of BI tools through bibliometric studies. | Mapping global trends in BI with bibliometric tools and analytics. | Critical reviews and bibliometric mapping of global trends. | Establishing global standards for BI interoperability and ethical AI usage. | educational data [56].<br>5. Hayward, D.V. et al. (2022). Exploring preservice teachers engagement with live models of universal design for learning and blended learning course delivery [57].<br>1. Khalil, M. et.al. (2023). Video analytics in digital learning environments: Exploring student behaviour across different learning contexts [58].<br>2. Junge, H. et.al. (2023). First steps of learning analytics in a blended learning general practice curriculum at Saarland University – A quantitative approach [59].<br>3. Algarni, A. et.al. Predicting at-risk students in higher education [18].<br>4. Macfadyen, L.P. et.al. (2023). A novel online genomic counseling and variant interpretation certificate: Learning design, learning analytics, and evaluation [60]<br>5. Llauro, A. et.al. (2023). Identification and comparison of the main variables affecting early university dropout rates according to knowledge area and institution [61]. | Bibliometric tools, Systematic reviews, Advanced learning analytics. |

Source: Elaborated by the authors.

future research [65,66]. The analysis uses scientific mapping techniques with Bibliometrix R and Biblioshiny as tools [67,68], providing an effective view of advances, current trends, and future directions. The research questions guiding this study [69,70] are as follows.

Main research question (RQ): How have the topics, scientific production, and methodological approaches evolved in BI research and data analysis in the academic management of HEIs?

The following specific questions helped to break down and answer the main question formulated:

RQ1: What are the main and relevant themes that have guided the evolution of the focus on BI and data analytics in the academic management of higher education institutions over the last five years?

RQ2: What are the most relevant academic sources in the field of BI and data analytics measured through their impact and importance according to the number of publications, h-index, and total citations?

RQ3: Who are the most influential authors in BI and data analytics and how do their contributions stand out in terms of productivity and impact?

RQ4: How is geographically distributed BI and data analytics research and what trends emerge when comparing different countries and continents?

RQ5: What are the main themes and trends identified in BI and data analytics research based on the analysis of the most cited articles worldwide?

RQ6: How do the educational sector interact with BI and data analytics in various con-texts?

RQ7: What are the predominant methodological approaches in BI and data analytics research and how do they relate to the thematic areas addressed?

Finally, the article was structure as follows: Section 2 comprehensively elucidates the methodology applied and the techniques employed for the study design, data collection, analysis, visualization, and interpretation in this bibliometric research. Section 3 presents the results of the bibliometric analysis. Section 4 comprehensively analyzes and discusses the evolution, topics and relevant research methodologies of BI and data analysis in the academic management of higher education institutions, including the main implications and limitations derived from

the exploration of the scientific literature. Finally, Section 5 summarizes the conclusions of this study and provides implications for future research initiatives.

2. Materials and methods

This research included a methodological procedure incorporating five stages for the overall progress of scientific mapping: study design, data collection, data analysis, analysis, visualization, and interpretation of data, specified by [71] and [72], and adaptations of [67,69,73], and [74], as illustrated in Fig. 1.

2.1. Study Design

The first phase of the research focused on study design, with a comprehensive analysis of various academic publications collected between 2019 and December 2023. The main objectives were to understand the evolution of BI and data analytics in academic management in higher education and to examine its relevance. In addition, we analyzed the influence of academic sources, identified the most prominent authors, explored the most relevant topics, and examined the relationship between BI, data analytics, and the higher education sector. This analysis also examined the methodological approaches used in the research, with the goal of identifying future lines of research. To achieve this, a comprehensive literature review and rigorous bibliometric analysis were integrated [75–77] with significant contributions from experts in the field. This methodological approach strengthened the research structure and provided a comprehensive view of the evolution and current state of BI and data analytics in higher education.

2.2. Data collection

In the second phase, the scientific database to consult was determined by comparing the two most relevant databases worldwide: Clarivate Analytics' Web of Science (WoS) and Elsevier's Scopus [78]. Scopus, which is widely recognized, contains >27 million abstracts, making it the most comprehensive database. Burnham (2006) noted that



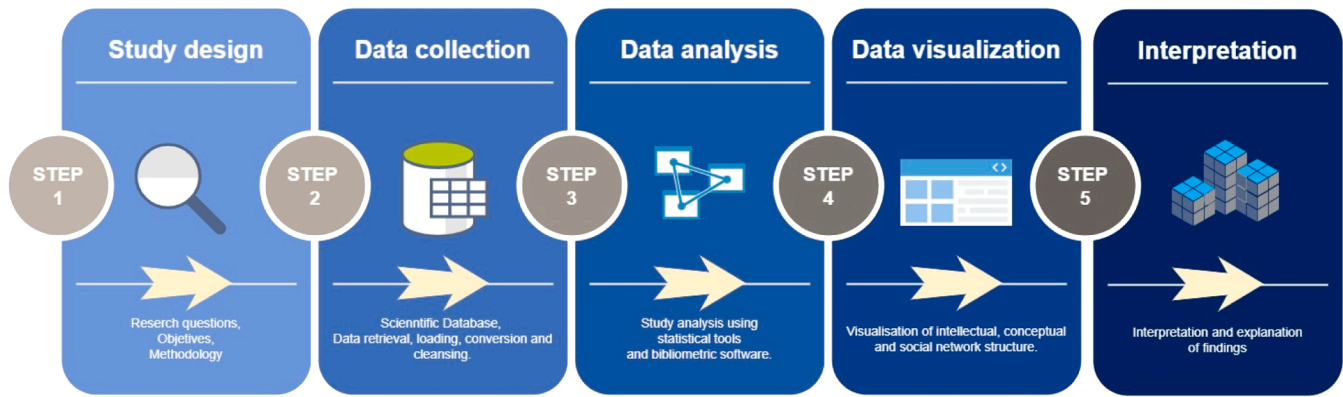


Fig. 1. Methodology steps, adapted from [69,73].

Scopus covers 97 % of all publications in WoS [79], and the inclusion of Scopus-exclusive journals may reduce the average impact and citation coverage in some countries [80,81]. Therefore, this study relied solely on Scopus data to cover a broader set of relevant journals [82], thus seeking a complete representation of the relevant scientific literature.

The guidelines for the search criteria included the fields of title, abstract, and author keywords [59]. Relevant keywords selected to facilitate the initial search. These keywords was include: "business intelligence" OR "business analytics" OR "academic analytics" OR "educational data mining" OR "educational big data" AND "higher education\*" OR "university\* OR college learning analytics. The choice of these terminologies based on the objective of achieving greater precision, speed, and focus, while seeking to incorporate a diversity of perspectives adapted to heterogeneous audiences. After a series of references, the documents containing the terms divided into three components: (1) Business Intelligence, (2) Data Analytics, (3) Academic Analytics, and (4) Higher Education. The adoption of these terminologies has led to an expanded collection of records that meet the objectives of this research. Table 2 lists the criteria used for the filtering process. This approach helps to obtain the most relevant results and to explore various aspects of BI and data analysis in the field of higher education. The main research yielded 2653 records between 1999 and December 2023.

Subsequently, additional filters have been used to improve the results [67,69]. The scope of the search was limited to the period from 2019 to December 2023, and generated 1623 records, as it was imperative to include recent literature. Furthermore, the research was limited to documents classified as articles (767) and/or reviews (19), disregarding editorials (4), books (12), book chapters (67), errata (8), data papers (2), retracted articles (2), conference reviews (54), and session papers (688), resulting in 786 records. The extracted data were cleaned using RStudio to eliminate duplicate entries and standardize terminology. Keywords were harmonized to unify concepts such as "academic analytics" and "learning analytics," thereby ensuring consistency and

minimizing redundancy. Furthermore, incomplete records were identified and excluded after the verification. These methodological steps ensured the integrity of the analyzed data.

No exclusions made based on journal category or specialization, which allowed the research to cover general and specialized journals simultaneously [70,83]. The inclusion criteria ensured that only articles and reviews published in English between 2019 and 2023 were considered, focusing on studies addressing at least one of the following key areas: Business Intelligence, Data Analytics, Academic Analytics, or Higher Education. Documents lacking abstracts, irrelevant content, or duplicates were excluded. Fig. 2 illustrates the systematic filtering process conducted using the PRISMA framework, delineating the sequential steps of identification, screening, and inclusion. Finally, only articles written in English (35) chosen, yielding 755 documents relevant to the study (Fig. 2). To improve the reliability of the study, we openly present bibliometric extraction in the Zenodo repository [84].

Accordingly, the academic team obtained bibliometric data from the Scopus platform in various file formats containing complete records and referenced sources such as RIS, BibTeX, and CSV. To make informed decisions, the team uses the Mendeley tool to select relevant documents. The processing and analysis of these files conducted extensively using the R statistical software, which is open-source, along with the Bibliometrix package. The extracted data cover specific details: (1) descriptive information, (2) evaluation of sources, (3) scrutiny of authors and citations, (4) examination of networks and keywords, and (5) geographical spread of articles. These data will used to draw relevant conclusions and findings.

2.3. Data analysis

The third stage of the research process focused on the choice of a unit of data analysis within the field of bibliometric, seeking to extract significant insights and understand the relationships between different variables in depth. To this end, the study applied multiple metrics and statistical techniques, such as the h-index, the total number of citations received, analysis and counting of documents according to their sources and authors. Within the framework of this research, Microsoft Excel was used for the visualization and presentation of data corresponding to research questions RQ1 to RQ3, complementing the analysis performed with Bibliometrix. Additionally, for the analysis of the most cited documents (RQ4), and the examination of the predominant methodological approaches in the research (RQ5), these tools used exclusively.

On the other hand, the Bibliometrix package and the Biblioshiny tool used to delve into aspects such as thematic evolution through author keyword analysis (RQ1), production trajectory of sources over time (RQ2), and author productivity according to Lotka's Law (RQ3). For questions RQ5, a "keyword analysis" technique implemented using Keywords Plus, Keywords' Authors, Keywords Title, and Keywords

Table 2  
Criteria for the filtering process.

| Criteria          | Value                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Topic             | (TITLE-ABS-KEY ("business intelligence" OR "business analytics") OR TITLE-ABS-KEY ("academic analytics" OR "learning analytics" OR "educational data mining" OR "educational big data") AND TITLE-ABS-KEY ("higher AND education*" OR "universit*")) AND PUBYEAR > 2018 AND PUBYEAR < 2024 AND (LIMIT-TO (DOCTYPE , "ar") OR LIMIT-TO (DOCTYPE , "re")) AND (LIMIT-TO (LANGUAGE , "English")) |
| Categories        | All categories without imposing restrictions on research.                                                                                                                                                                                                                                                                                                                                     |
| Type of documents | Articles, review articles.                                                                                                                                                                                                                                                                                                                                                                    |
| Language          | Only articles in English.                                                                                                                                                                                                                                                                                                                                                                     |
| Time frame        | 2019 to December 2023.                                                                                                                                                                                                                                                                                                                                                                        |

Source: Elaborated by the authors.

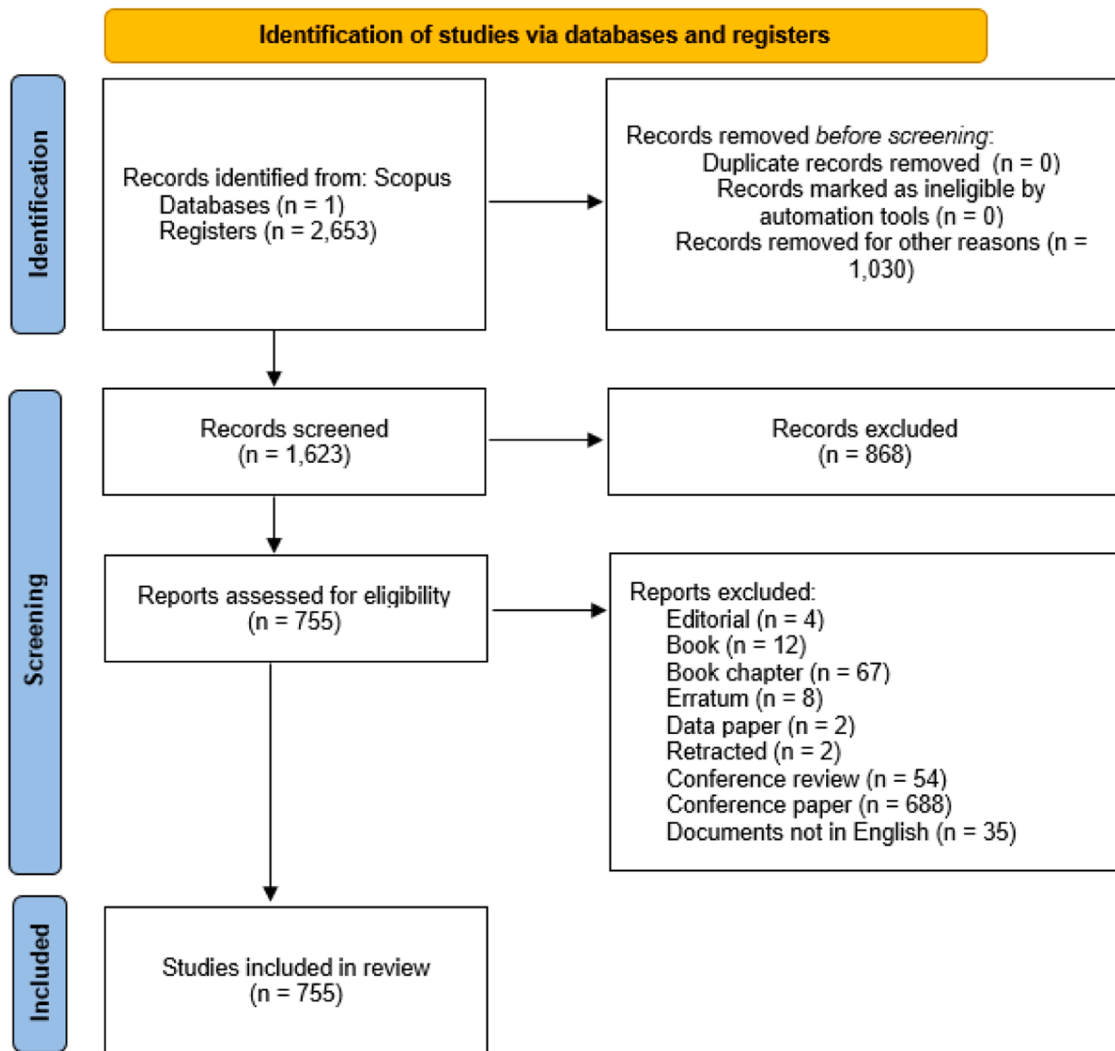


Fig. 2. Filtering process of scientific literature. Source: Elaborated by the authors in [85].

Abstract. This conceptual analysis approach [73] facilitated the identification of the frequency of keyword usage in the analyzed documents, revealing connections between BI, data analytics, and the higher education sector. In addition, this method made it possible to discover the predominant methodological approaches and visualize future lines of research through the graphical representation of interconnected thematic clusters, thus simplifying the interpretation of research areas [86].

To carry out this analysis efficiently, we used the following: (1) the RStudio software package (R version 4.3.0, <https://www.R-project.org>, January 15, 2024), (2) Bibliometrix R package (version 4.2.1, <http://www.bibliometrix.org>, accessed January 15, 2024), and (3) Biblioshiny application (version 1.7.5), which incorporates scientometric tools programmed in the R language. These tools were selected based on their capacity to process large datasets, compute advanced bibliometric indicators, such as the h-index, and generate thematic clusters and keyword mappings. Biblioshiny's interactive interface facilitated dynamic data exploration, whereas RStudio ensured reproducibility and robust statistical analysis. Furthermore, the Microsoft Excel 2016 desktop application was utilized for manual cross-verification and descriptive data representation, complementing the advanced analytical capabilities of RStudio and Biblioshiny. Collectively, these algorithms provide efficient statistical analysis and integrated visualization tools, enabling a comprehensive understanding of the research field [67]. While these tools provide advanced capabilities for bibliometric analysis and visualization, they also possess certain

limitations. For instance, RStudio and Biblioshiny require a steep learning curve and proficiency in statistical coding, which may present challenges for researchers lacking prior experience. Furthermore, Microsoft Excel, although valuable for manual verification and basic data management, lacks the sophisticated bibliometric functionalities required for large-scale analysis. Acknowledging these constraints ensured a comprehensive understanding of the methodological approach.

#### 2.4. Data visualization

In the fourth phase of the study, specific visualizations assigned to each research question to illustrate the results: RQ1 represented by a line graph with markers, highlighting temporal evolution. For RQ2, RQ3, and RQ4, we produced tables in RStudio to compare and analyze the data. RQ5 represented by sets of word clouds to focus on the frequency of specific terms. Each visualization carefully chosen to highlight the key aspects of the answers to the research questions, facilitating a clear interpretation of the data collected.

#### 2.5. Interpretation

The final phase of the research involves an investigative and explanatory review (Oliveira et al., 2022), which illustrates how the findings were analyzed and interpreted according to the research

objectives and questions. The analysis highlights various aspects of this specialized field, covers the entire period of publication, and describes salient, fundamental, specialized, and emerging themes, along with their evolution over time. To interpret the results, we employed bibliometric techniques and relied on the extensive experience of researchers in the field, although these methods are not a substitute for a comprehensive literature review. Software-assisted processes were used in stages two–four [67]. This paper concisely presents the conclusions and discusses the implications and significance of the findings. This study also provides suggestions for prospective research.

3. Results

The growing interest of the scientific community in BI-related research and data analytics plays a crucial role in the academic governance of education institutions, as observed in the relevant literature. The analysis presented in Table 3 provides basic information derived from Bibliometrix and Biblioshiny on the topic discussed in 755 articles, demonstrating the aforementioned upward trend in scientific output between 2019 and December 2023. Our research includes a full historical description of the annual publications of all selected articles.

Within the specified study period, the research identified 755 manuscripts, including 736 articles (97.5 %) and 19 reviews (2.5 %). At the level of higher education institutions (HEIs), researchers collected data from 343 scientific journals related to BI and data analysis. In 2023, the average number of annual publications reached 9.9, indicating that the current interest and recognition align with this field of study. However, 2023 witnessed the most substantial increase in scholarly output, peaking at 178 publications. This evidence suggests that there is an important body of relevant information within the scientific community

Table 3  
Main information.

| Type                            | Description                                                                             | Results             |
|---------------------------------|-----------------------------------------------------------------------------------------|---------------------|
| Main Information about data     |                                                                                         |                     |
| Period                          | Years of publication.                                                                   | 2002: December 2023 |
| Sources                         | The frequency distribution of sources as journals.                                      | 343                 |
| Documents                       | Total number of documents.                                                              | 755                 |
| Annual Growth Rate %            | Average number of Annual Growth                                                         | 9.9                 |
| Document Average Age            | Number of the average age of the document.                                              | 2.86                |
| Average citations per document  | Average total number of citations per document.                                         | 11.57               |
| References                      | Total number of references or citations.                                                | 31,156              |
| DOCUMENT CONTENTS               |                                                                                         |                     |
| Keywords Plus (ID)              | Total number of phrases that frequently appear in the title of an article's references. | 2029                |
| Author's Keywords (DE)          | Total number of keywords                                                                | 2035                |
| AUTHORS                         |                                                                                         |                     |
| Authors                         | Total number of authors.                                                                | 2345                |
| Authors of single-authored docs | The number of single authors per articles.                                              | 61                  |
| AUTHORS COLLABORATION           |                                                                                         |                     |
| Single-authored document        | The number of documents written by a single author.                                     | 66                  |
| Co-Authors per document         | Average number of co-authors in each document.                                          | 3.7                 |
| International co-authorships %  | Average number of International co-authorships.                                         | 22.65               |
| DOCUMENT TYPES                  |                                                                                         |                     |
| Article                         | The number of article.                                                                  | 736                 |
| Review                          | The number of articles of review.                                                       | 19                  |

Source: Adapted summary table from Biblioshiny.

on this topic.

The manuscripts have an average age of 2.86 years, indicating the recent nature of most of these publications. This serves as an indicator of the continuous development of the BI field and data analysis, which requires consistent research and updates. Each manuscript obtained, on average, 11.57 citations, exemplifying its high interest and recognition within the academic community. These designations served to assess their importance and impact on the field of BI and data analytics in higher education institutions.

In relation to references, the study identified 31,156 citations in manuscripts related to BI and data analysis in higher education institutions (HEIs). This indicates the presence of a strong literature and research base that supports and validates the research in this area. Citations also provide an opportunity to further explore and examine this topic. In articles related to the variable under analysis, the research discovered 2029 additional keywords (ID) and 2035 author keywords (ED). These keywords provide a more complex and precise perspective on the topics and methodologies addressed in publications on BI and data analytics in higher education institutions.

Regarding authors, 2345 academics have contributed to manuscripts related to BI and data analysis in institutions of higher education. This demonstrates the wide variety and participation of the scientific community engaged in research on this topic. The collaboration index (CI) reached a value of 3.7, which represents the average number of authors per jointly authored or co-authored article [87]. Within this average, approximately 22.65 authors collaborate internationally, demonstrating a substantial level of global cooperation in research in this field. This highlights the importance of collaboration between researchers from different countries. Overall, there has been a notable increase in publications between 2019 and 2023 (Fig. 3). Over the past five years, there has been growing interest in BI research and data analytics in higher education institutions from the scientific community. This emphasizes the importance of this topic in an increasingly digitized world and the need to continue researching and updating knowledge in this evolving field. As a result, the research obtained 755 articles from Scopus.

3.1. Evolutionary Analysis of BI and data analytics: main and relevant themes in higher education (RQ1)

The alluvial graph in Fig. 4 shows a longitudinal analysis of a thematic map derived from the keywords of the author in the manuscript database. This analysis allowed us to observe the evolution of topics related to BI and data analysis in higher education institutions (HEIs). By dividing the time into intervals, we can study and identify important trends and issues. Considering the distribution of publications by year, we divided our collection into three periods using two cutoff points: 2020 and 2022. The parameters of these frames were as follows: number of words=250, Min Cluster Frequency (per thousand docs) =5, Weight index = Inclusion Index weighted by Word-Occurrences; Min Weight Index=0.1, Number of labels (for each cluster) =3, and clustering Algorithm=Walktrap [88].

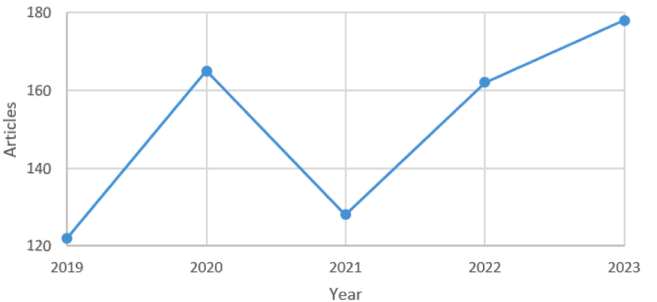


Fig. 3. Annual scientific production. Source: Authors' elaboration using the R-bibliometrix package, Biblioshiny and MS Excel.

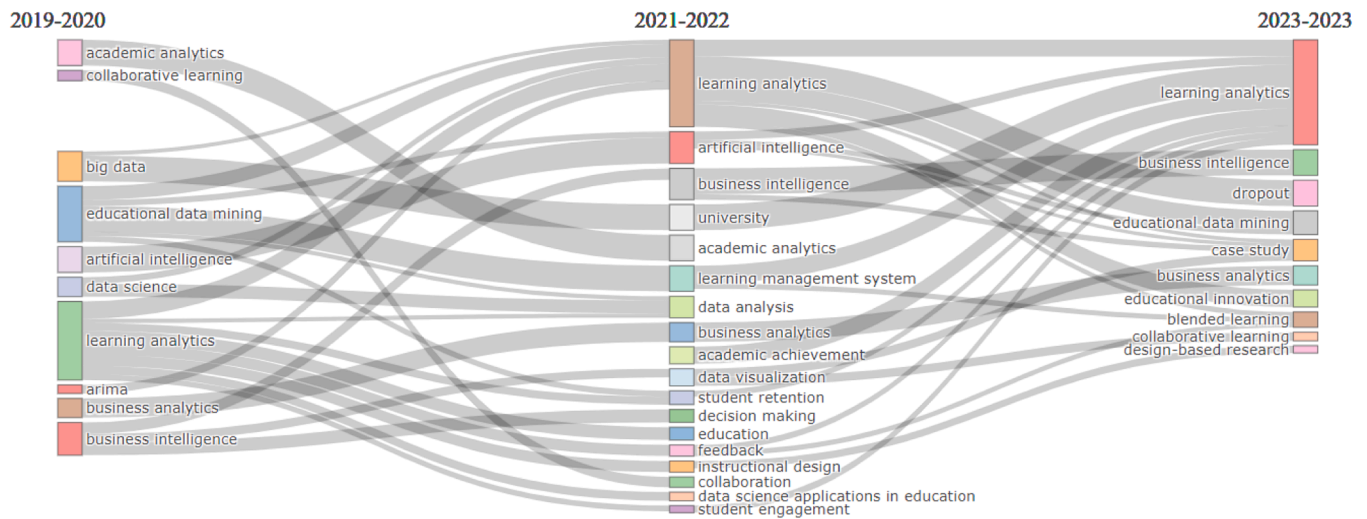


Fig. 4. Thematic evolution from 2019 to 2023. Source: Scopus database. Authors' elaboration using the R-bibliometrix package, Biblioshiny.

From 2019 to 2020, research has focused on several topics, including learning analytics, educational data mining, business intelligence, big data, and academic analytics. From 2021 to 2022, emphasis shifted to learning analytics, artificial intelligence, business intelligence universities, and academic analytics. The most relevant thematic trends among researchers in the last quarter of 2023 are learning analytics, business intelligence, dropout, educational data mining, and business analytics. It is possible to note that certain topics have shown a tendency to merge, split into subtopics or even disappear over time. For example, academic analytics, data analysis, and decision-making were visible themes between 2019 and 2022 but have since disappeared.

Fig. 5 illustrates the progression of topics within the study domain. The author's keyword network is subject to a clustering algorithm that allows the discernment, description, and accentuation of numerous themes within a particular area. Each cluster or theme is described through a thematic or specific diagram [89] by period, which accentuates the importance (centrality) and advancement (density) of the thematic network.

During the period 2019–2020 (Fig. 5a), topics such as educational data mining (304 papers), learning analytics (260), business intelligence (29), big data (38), business analytics (10), data science (8), and outcome-based education (5). A total of 649 papers focused on predicting academic performance using data mining and learning analytics. These studies explored student behavior, interaction data in virtual environments, and demographic characteristics to develop predictive models and provide early interventions. Among the most cited articles in 2019, Francis B.K. and Babu, S. S., highlighting the importance of data mining to predict academic performance [90]; and, Abu Zohair, Lubna Mahmoud, further considers supporting at-risk students, improving educational resources and university reputation [65]. In 2020, Mengash, H. A. emphasizes the relevance of pre-admission criteria, such as high school GPA, in accurately predicting student achievement [38]. Hooshyar, Pedaste, and Yang proposed a PPP algorithm based on task delivery behavior, achieving 96 % accuracy [46].

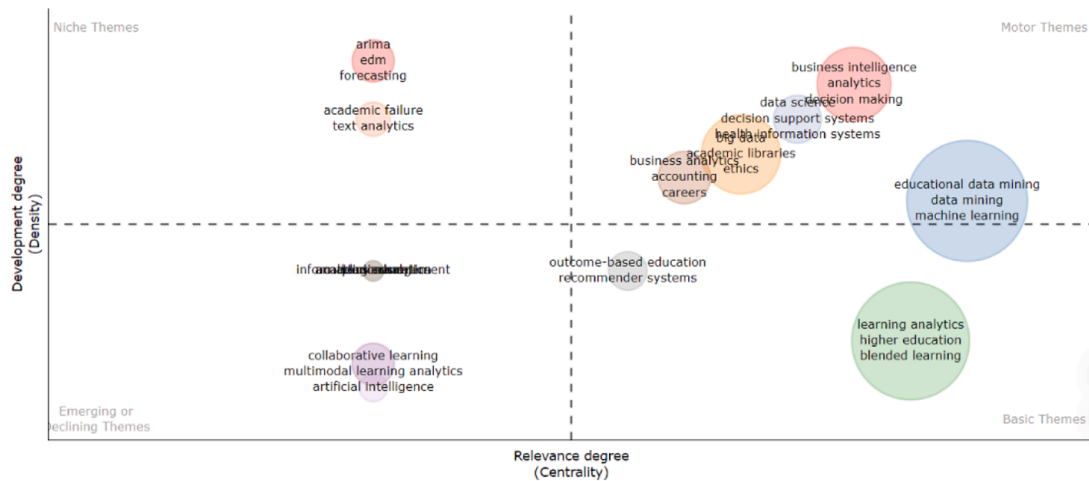
In the period 2021–2022 (Fig. 5b), learning analytics continued to be studied (472 papers), followed by new topics, such as feedback (21), student retention (13), student engagement (18), and data analysis (12). This particular year contributed a considerable number of 536 manuscripts, encompassing a broad spectrum of subjects such as employing machine learning techniques to anticipate student performance, examining student dropout rates in higher education, applying learning analytics to prevent dropout, assessing the influence of the pandemic on online learning, and constructing predictive models using diverse statistical methodologies and machine learning algorithms. Among the

most cited articles in 2021, Albreiki, B., Zaki, N. and Alashwal, H. highlight the crucial role of Educational Data Mining (EDM) to improve learning environments, they use machine learning to forecast student dropout, address challenges in this area and highlight the essential contribution of machine learning to student achievement [19]. The following year, 2022, Yağcı, M. proposes a model based on machine learning algorithms to predict final grades of university students, highlighting an accuracy of 70–75 % [91].

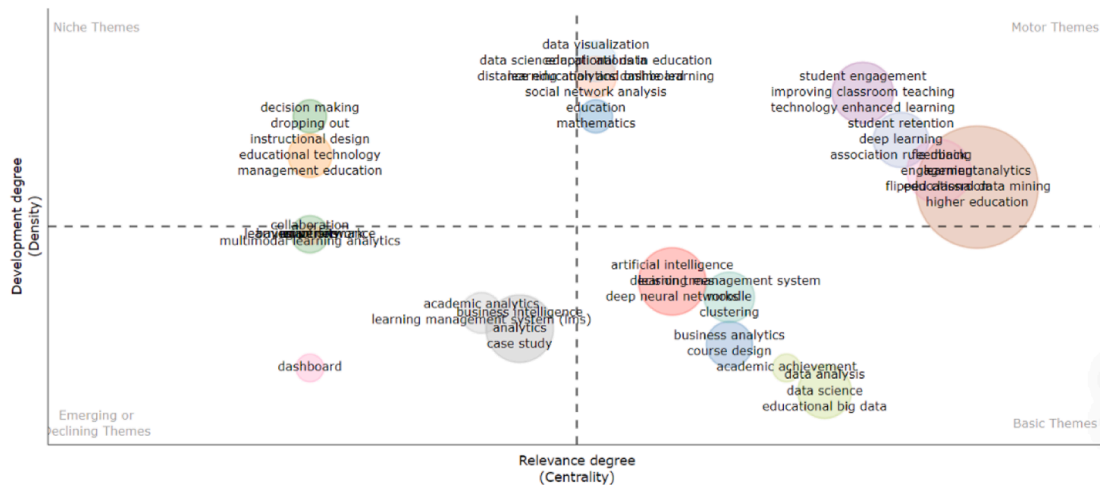
In 2023 (Fig. 5c), learning analytics (157 papers), of which 23 were specific to higher education, demonstrated the interest of researchers in opening new frontiers in this field, as well as educational data mining (169), followed by design-based research (10), blended learning (10), educational innovation (3), business analytics (12), virtual learning environments (14), and business intelligence (8), among others (7). At the end of the last quarter, 375 manuscripts focused on studies that addressed learning analytics, predictive modeling, educational data mining, and machine learning to improve teaching, student performance, and dropout prediction. Among the most-cited studies, Thili et al. proposed a non-intrusive method for modeling student personalities in an intelligent learning environment using a learning analytics approach [92], whereas Zhang Y. used a deep learning model to predict and understand student performance using learning log data, highlighting the importance of detailed interpretations [93].

In summary, the number of published manuscripts has increased in recent years. This analysis emphasizes the importance of understanding and developing further BI and data analytics studies in various contexts in the current digital era. In the last section of thematic evolution in 2023 (Fig. 5c), the topics with the highest number of occurrences were learning analytics and higher education, which are also driving topics with higher relevance in this field. On the other hand, educational data mining, machine learning, business analytics, dropout, data analytics and educational innovation are basic and transversal themes in this field. Business Intelligence is currently on the border between basic but emerging topics. This study explores several themes per year and presents remarkable insights into the areas of focus and patterns in academic research in the field of BI and data analytics in academic administration in higher education institutions. These findings have the potential to establish a solid foundation for future research and contribute to the formulation of more effective strategies in a variety of contexts.

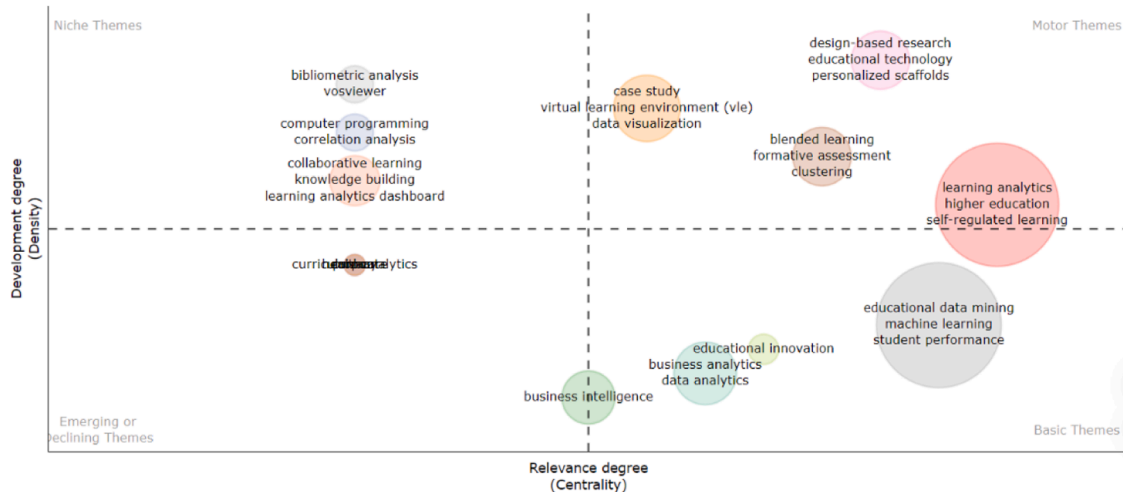




(a) Thematic evolution map period 2019–2020.



(b) Thematic evolution map period 2021–2022.



(c) Thematic evolution map period 2023.

**Fig. 5.** Thematic evolution. (a) 2019–2020. (b) 2021–2022. (c) 2023 Source: Scopus database. Authors' elaboration using the R-Bibliometrix package, Biblioshiny.



### 3.2. Analysis of the local impact and importance of sources: relevant journals (RQ2)

We analyzed 755 peer-reviewed published articles from the Scopus database. The distribution of the articles (Table 4) shows a list of the 10 main specialized journals in this study. All these studies focused on the areas of learning analytics, educational technology, and higher education. Their content covers research, advances, and applications related to the use of technology in education, data analytics to improve learning, and the impact of technology in educational environments. Topics, such as interactive learning, emerging technologies, access to education, computing applied to education, and data analysis in educational contexts, are common in these publications. Table 4 shows the order of the numbers of publications and citations.

The graph in Fig. 6 shows the frequency of distribution of journals centered on the number of publications, where the journal Applied Sciences (Switzerland), with 20 manuscripts, is the highest, and the lowest is the journal Internet and Higher Education with seven manuscripts. An inspection of the data from 2019 to 2023 revealed a remarkable increase in the number of publications on the subject. However, the graph illustrates the results of the local Loess regression, which integrates variables such as the number and time of publication of the analyzed journal. This approach allows for an unbounded distribution function, where values less than zero assumed if the data are close to zero. Its use results in superior visual output and emphasizes the publication period intervals [94].

Another analysis revealed significant changes when evaluating journals solely by their impact factor, that is, the average number of citations per article over a 2-year period. Under this criterion, the journal rankings shifted. The International Journal of Educational Technology in Higher Education, with an impact factor of 8.6 and a Q1 classification, would take the first position. Closely following, Internet and Higher Education, also with an impact factor of 8.6, and a Q1 classification, would be in the second position. Followed by British Journal of Educational Technology, Education and Information Technologies, Interactive Learning Environments, all of them Q1 category and with an impact factor between 6.6 and 5.4 as of 2022, according to Clarivate's Journal Citation Report (JCR). However, using the impact factor alone may favor journals with few highly cited articles.

The h, g, and m indexes balance production and citations for a more comprehensive view. Where, the Journal of Learning Analytics (Q1, Impact Factor=3.9, JCR metrics) would be the top ranked source for literature production in BI and data analytics, with an h-index of 11 and a g-index of 13, its 20 publications have 472 total citations, the first article was published in 2019 [95,96]. The journal Applied Sciences (Switzerland) (Q2, Impact Factor=2.7, JCR metrics) is the second-best positioned in the list, with an h-index of 10 and a g-index of 17, but the first by number of publications (22) with 310 citations; its first

published article was in 2019 [97–99]. Education and Information Technologies ranks third; it publishes research on the use of ICT in education. It has an impact factor of 5.5, 9 h-index, 19 g-index and 1.5 m-index and 400 citations.

However, for this study we consider evaluating all the bibliometric indicators contained in Table 4 ordered by the "Rank" column, i.e., evaluated by: impact factor, h-index, m-index, g-index, number of publications and citations in a comprehensive manner, where the top 5 scientific journals are: 1) British Journal of Educational Technology (Q1): Leading journal in the field of educational technology. It has an impact factor of 6.6, 9 h-index, 15 g-index and 1.5 m-index. It publishes research on the innovative use of technology in education [55, 100–104], 2) Education and Information Technologies (Q1): It publishes research on the use of ICT in education. It has an impact factor of 5.5, 9 h-index, 19 g-index and 1.5 m-index [15,21,105–107], 3) Journal of Learning Analytics (Q1): Focuses on the emerging field of learning analytics. It has an impact factor of 3.9, 11 h-index, 20 g-index and 1.83 m-index [7,95,108–111]; and 4) IEEE Access (Q2): Open Access Journal covering Topics in Electrical Engineering and Computer Science. It has an impact factor of 3.9, 9 h-index, 18 g-index, and 1.5 m-index [112, 113]; and 5) Interactive Learning Environments (Q1): Research on the design and use of interactive learning environments. It has an impact factor of 5.4, 8 h-index, 10 g-index and 1.33 m-index [45,92,114,115].

### 3.3. Analysis of the most influential authors in the study: productivity and impact (RQ3)

#### 3.3.1. Author productivity

Lotka's law, initially introduced in 1926 as a mathematical formulation, aims to elucidate the publication frequency of authors in a specific research field [116]. This law postulates that the number of authors who contribute to research in a given period is the proportion of those who generate an individual contribution [67,116]. Its mathematical representation is expressed as follows:  $x^n \cdot y_x = C$ , where  $y_x$  represents the number of authors writing  $x$  articles in a particular field of research [76], and is inversely proportional to the square of  $n$ . The constants  $C$  and  $n$  can be approximated using calculations. The results presented in Fig. 7 are aligned with Lotka's findings, indicating that authors produce an average of two publications in a given research field. Furthermore, the figure illustrates the percentage of authors engaged in this research field, suggesting that it is young and expanding in nature. Specifically, out of a total of 2345 authors derived from the object of study, approximately 89 % have published a single research article, while only 7.5 % have authored two scientific articles, 1.7 % have published three articles, and 0.6 % have published four to five published articles.

#### 3.3.2. Author Impact

The following segment presents the most influential scholars in the

**Table 4**  
Top ten sources with impact and relevance.

| Source                                                              | Category | Impact Factor (JCR) | h_index | g_index | m_index | TC  | NP | PY_start | Rank |
|---------------------------------------------------------------------|----------|---------------------|---------|---------|---------|-----|----|----------|------|
| (in 2022)                                                           |          |                     |         |         |         |     |    |          |      |
| Applied Sciences (Switzerland)                                      | Q2       | 2.7                 | 10      | 17      | 1.67    | 310 | 22 | 2019     | 5    |
| Journal of Learning Analytics                                       | Q1       | 3.9                 | 11      | 20      | 1.83    | 472 | 20 | 2019     | 3    |
| IEEE Access                                                         | Q2       | 3.9                 | 9       | 18      | 1.50    | 340 | 20 | 2019     | 4    |
| Education and Information Technologies                              | Q1       | 5.5                 | 9       | 19      | 1.50    | 400 | 19 | 2019     | 2    |
| International Journal of Emerging Technologies In Learning          | Q2       | –                   | 6       | 12      | 1.00    | 171 | 19 | 2019     | 9    |
| International Journal of Advanced Computer Science and Applications | Q4       | 0.9                 | 6       | 9       | 1.00    | 94  | 18 | 2019     | 10   |
| British Journal of Educational Technology                           | Q1       | 6.6                 | 9       | 15      | 1.50    | 382 | 15 | 2019     | 1    |
| International Journal of Educational Technology in Higher Education | Q1       | 8.6                 | 7       | 10      | 1.17    | 336 | 10 | 2019     | 7    |
| Interactive Learning Environments                                   | Q1       | 5.4                 | 8       | 10      | 1.33    | 262 | 10 | 2019     | 6    |
| Internet and Higher Education                                       | Q1       | 8.6                 | 7       | 7       | 1.17    | 244 | 7  | 2019     | 8    |

Source: Elaborated by the authors.

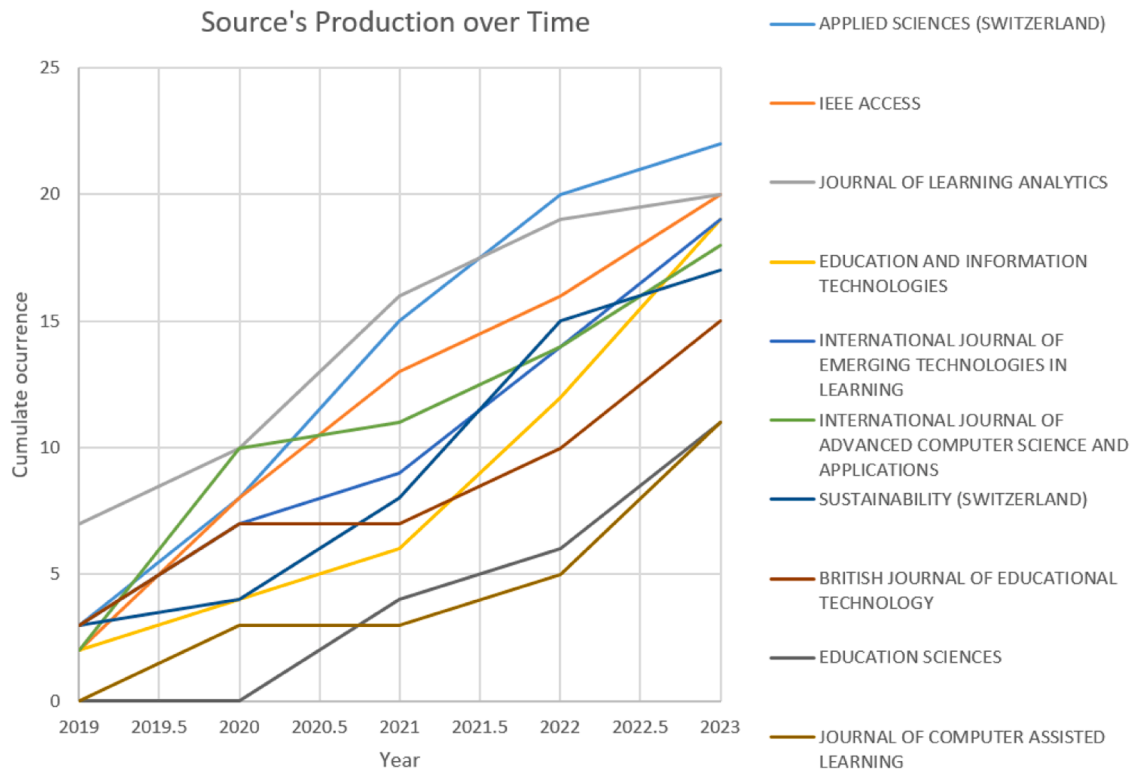


Fig. 6. Source production over time. Source: Authors' elaboration using the R-Bibliometrix package, Biblioshiny, and MS Excel.

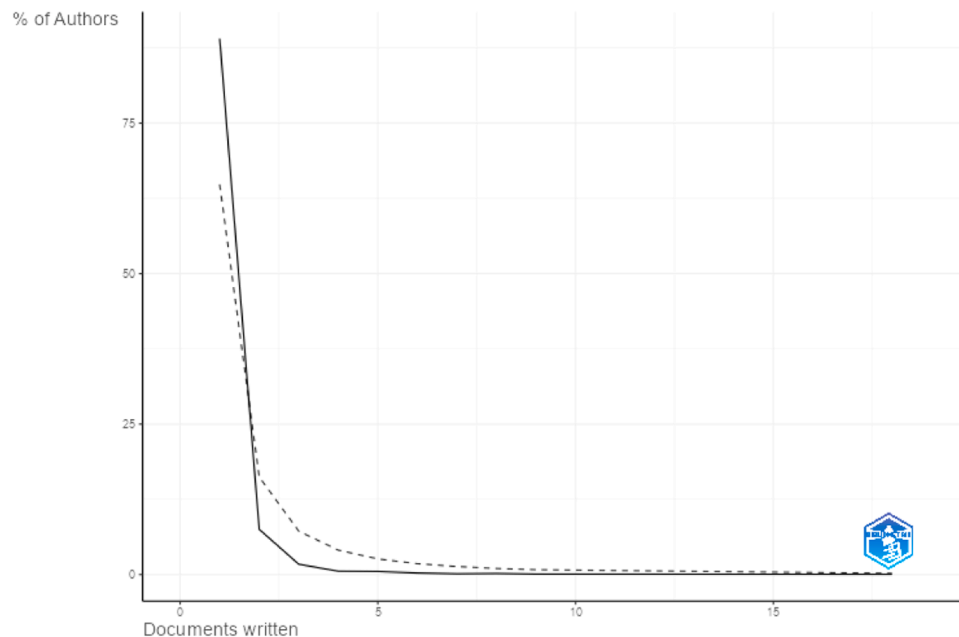


Fig. 7. Lotka's Law. Source: Authors' elaboration using the R-bibliometrix and Biblioshiny package.

field of BI and data analytics in higher education institutions (HEIs). Table 5 provides a compilation of academics and their work, focusing on the top 10 individuals in the hierarchy. This list emphasizes three fundamental elements ordered by the number of publications, h-index, and total citations (TC) obtained by each author.

At first glance, the researcher with the highest number of published papers is Nuankaew, P. from the University of Phayao, who has written 18 publications in total [117]. Despite having the highest number of articles, the h-index is only five, which means that at least five of his

papers received citations five times or more. In addition, his research received 89 citations, which is the lowest among the authors.

Herodotou, C., exhibits an h-index of seven, a compilation of 9 published articles, and 316 citations, which are the most notable indicators among all authors. This makes him the author with the highest impact on productivity [7,96,100,118,119]. Consequently, we can deduce that the works of Herodotou, C., a researcher at The Open University, enjoy wide recognition and usage in research related to BI and data analysis in higher education institutions.

**Table 5**  
The 10 most influential authors.

| Authors                        | Institution                        | Country     | Articles | h_index | TC  |
|--------------------------------|------------------------------------|-------------|----------|---------|-----|
| Nuankaew, Praty                | University of Phayao               | Thailand    | 18       | 5       | 89  |
| Herodotou, Christothea         | The Open University                | UK          | 9        | 7       | 316 |
| Rienties, Bart                 | The Open University                | UK          | 8        | 7       | 302 |
| Tsai, Yi Shan                  | Monash University                  | Australia   | 8        | 7       | 198 |
| Gašević, Dragan                | Monash University                  | Australia   | 8        | 5       | 123 |
| Whitlock-Wainwright, Alexander | Monash University                  | Australia   | 7        | 6       | 149 |
| Boroowa, Avinash               | The Open University                | UK          | 6        | 6       | 283 |
| Drachsler, Hendrik             | Open University of the Netherlands | Netherlands | 6        | 5       | 142 |
| Hlosta, Martin                 | The Open University                | UK          | 5        | 5       | 247 |
| Luján-Mora, Sergio             | University of Alicante             | Spain       | 5        | 5       | 141 |

Source: Authors' elaboration.

However, there are different authors (Rienties et al. [109,111,120]; Tsai, Y. S. [121,122]; Gašević, D. [123,124]), who have published eight articles each; the impact and influence of their respective contributions vary. Rienties, B. stands out with an h-index of 7 and 302 citations, being the second highest citation figure in the table. This means that their work is widely recognized and used in BI and data analytics studies on HEIs.

Whitlock-Wainwright [121,124,125], Boroowa, D. [126], Hlosta M. [100,120], and Luján-Mora S. [127,128] are authors who, despite having a moderate productivity of 5–6 published articles, show a high impact in the field of educational technology according to their bibliometric indicators. All have an h-index of 5–6 and a total number of citations between 141 and 283, which reveals that their contributions have a wide repercussion and influence among the scientific community in this field.

Fig. 8 presents a co-authorship network visualization that elucidates the collaborative dynamics in Business Intelligence (BI) and data analytics research within higher education. The visualization reveals denser clusters, such as those led by Herodotou and Rienties, which indicate robust collaborative networks in regions such as Europe, characterized

by a high concentration of publications and citations. Conversely, less interconnected clusters, exemplified by Villegas-Ch and Luján-Mora, suggest limitations in the integration of authors from underrepresented regions such as Latin America. This disparity underscores the necessity for initiatives that promote international collaboration and enhance the inclusion of regional perspectives, which are crucial for addressing specific educational challenges and enriching the field of study.

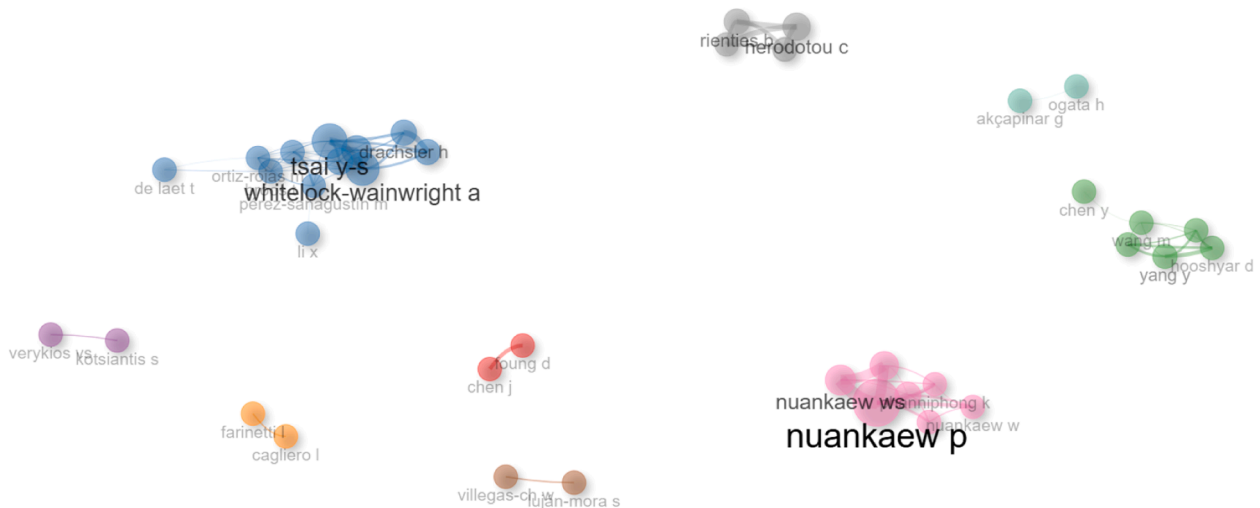
Furthermore, the structure of thematic clusters reflects the geographic and disciplinary trends. For instance, the pink cluster, led by Nuankaew, highlights Asia's specialization in predictive models and educational analytics, whereas European clusters focus on educational technologies and online learning. However, the disconnection between thematic clusters indicates that the relevant areas remain isolated, limiting interdisciplinary knowledge transfer. To address this fragmentation, it is imperative to foster global networks that connect researchers and topics, thereby promoting a more inclusive and collaborative approach to advancing BI and data analytics in higher education.

In general, although the number of articles indicates productivity, the data indicate that the h-index and total citations are better indicators of influence and impact in the field of BI and data analytics in HEIs. This positions them as relevant authors and references in the discipline, complementing Herodotou and Rienties, who are leaders in both productivity and impact. However, this vision is simplified, and more metrics and indicators are required to assess the relevance and impact of an author in a more complete manner.

3.4. Analysis of the distribution of publications by country and continent: exploration of geographic trends (RQ4)

In this section, we find information on the geographical distribution of publications, that is, where studies related to BI and data analytics published in HEIs. Table 6 shows the ten most important countries, together with the number of articles and citations on the subject. The United States has a total number of publications of 829, reflecting its prolific scientific output [129–131]. However, in total citations, although first, they only reached 297 citations, suggesting a lower relative impact. The United Kingdom ranks second in publications with 726 publications, showing a strong specialization in the subject.

However, it falls in fifth place in citations (147), indicating that the scope of its work is limited [118,132–134]. China has a high volume of publications (717) but few citations (230). This indicates productivity but the marginal influence of its research [114,135,136]. Spain balances publications (596) and citations (195) with solid metrics for both indicators. It is positioned as a relevant player [48,137,138]. Australia



**Fig. 8.** Collaboration Network. Source: Authors' elaboration using the R-bibliometrix and Biblioshiny package.

**Table 6**

Top countries in terms of citations and publications.

| Country        | Total Citations | Country        | Number of Publications |
|----------------|-----------------|----------------|------------------------|
| USA            | 297             | USA            | 829                    |
| China          | 230             | United Kingdom | 726                    |
| Australia      | 214             | China          | 717                    |
| Spain          | 195             | Spain          | 596                    |
| United Kingdom | 147             | Australia      | 428                    |
| Saudi Arabia   | 120             | Turkey         | 398                    |
| Italy          | 86              | Germany        | 358                    |
| Germany        | 79              | Italy          | 317                    |
| Thailand       | 74              | India          | 280                    |
| Canada         | 71              | Saudi Arabia   | 278                    |

Source: Authors' elaboration.

showed a similar pattern, with 428 articles and 214 citations. Its production and impact are proportional [121,124,125]. Turkey ranks sixth in number of publications, with 398 papers [58,139]. This indicates a strong interest in and production of the subject in that country.

However, since the top ten lack citation data, evaluating the impact of their contributions is not possible. Italy (317 publications) and Germany (358 publications) stand out for their high number of publications, suggesting that their contributions are numerous, although they are less influential in terms of citations. India, with 280 publications, also demonstrated active participation in BIs and educational data analyses. As in Turkey, no citation figures are available to measure influence. Saudi Arabia (278 publications but only 120 citations), the distance between its productivity and citations suggests limitations in influence that must be reversed. Thailand (74 citations) shows interest in the region but works with limited impact.

In summary, the geographical distribution of publications and citations in BIs and data analysis shows a clear dominance of the United States and China, with other countries contributing significantly but at different scales. Australia and Saudi Arabia had a relatively high proportion of citations compared to the number of publications. This could indicate that research from these countries has a significant impact on the scientific community despite having a smaller number of publications. The United States (USA), China, Spain, Ecuador, Saudi Arabia, Pakistan, Australia, Germany, Canada, Australia, the United Kingdom, and the Netherlands showed a remarkable degree of cooperation

between nations (Fig. 9), considering the number of references and a significantly high average number of references per article, suggesting strong collaboration and an emphasis on research and novelty in this area.

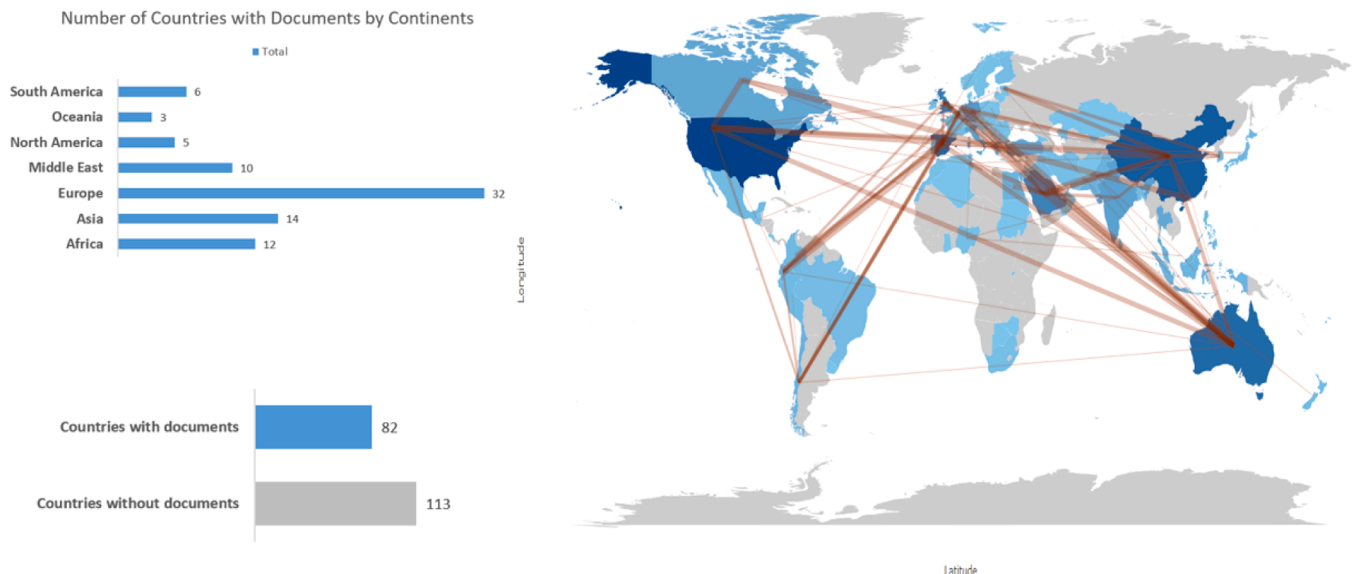
When analyzing the data by continent, Europe showed the highest participation in 32 countries, implying great enthusiasm and contribution to the field of BI exploration and data analysis. The abundance of educational establishments and research facilities in this region could explain this phenomenon. Asia also shows remarkable participation in 14 countries, signifying a growing interest in these research fields. Cultural heterogeneity and rapid technological advancement in Asian nations could be contributing factors to this trend. Africa and the Middle East showed moderate percentages, with 12 and 10 countries, respectively. This could reflect a growing interest in these regions, although investments and resources devoted to research may not yet be as significant as in other global sectors. Oceania and South America show lower proportions, with three and six countries, respectively. This observation could imply that the importance of research on business intellect and data analytics in higher education institutions is not as evident in these regions as in other areas.

Finally, after an exhaustive analysis of the dataset, it is evident that out of 195 countries [140], 82 countries have addressed the issue of BI and data analysis in the management of higher education institutions in their manuscripts. In contrast, other countries have not published 113 studies on this topic. The absence of contributions to BI and data analytics in these 113 nations highlights the urgency to boost research and knowledge dissemination in these areas. It is imperative that these nations share their perspectives and experiences in BI and data analytics, which could catalyze the exchange of knowledge and best practices at the international level. Overall, this analysis underscores the relevance of BI and data analytics as a global research topic, and highlights the importance of enriching geographic diversity in the production of knowledge in this area.

### 3.5. Analysis of the most cited papers in BI and data analytics in HEIs. Main topics and trends (RQ5)

By examining Table 7 of the top 10 most cited articles worldwide on BI and data analytics, we can identify several important trends and themes in current research in this area.

Overall, it is evident that the most popular articles in the three thematic areas focused on a) digital and academic entrepreneurship, b)



**Fig. 9.** Geographical distribution of publications among countries and continents. Source: Authors' elaboration using Biblioshiny and MS-Excel.



**Table 7**

Most cited documents.

| Rank | Most cited documents worldwide                                                                                                      | Authors of the paper                          | Total Citations | Ref. No. |
|------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------|----------|
| 1    | "Digital academic entrepreneurship: The potential of digital technologies on academic entrepreneurship"                             | Rippa, P., and Secundo, G. (2019).            | 189             | [141]    |
| 2    | "Utilising learning analytics to support study success in higher education: a systematic review"                                    | Ifenthaler, D., and Yau, J. Y. K. (2020).     | 129             | [9]      |
| 3    | "Learner satisfaction, engagement and performances in an online module: Implications for institutional e-learning policy"           | Rajabalee, Y. B., and Santally, M. I. (2021). | 120             | [40]     |
| 4    | "The efficacy of learning analytics interventions in higher education: A systematic review"                                         | Larrabee Sønderlund, A., et.al. (2019).       | 120             | [132]    |
| 5    | "Predicting Academic Performance of Students Using a Hybrid Data Mining Approach"                                                   | Francis, B. K., and Babu, S. S. (2019).       | 118             | [90]     |
| 6    | "Using data mining techniques to predict student performance to support decision making in university admission systems"            | Mengash, H. A. (2020).                        | 117             | [38]     |
| 7    | "Prediction of Student's performance by modelling small dataset size"                                                               | Abu Zohair, L. M. (2019).                     | 105             | [142]    |
| 8    | "The impact of engineering students' performance in the first three years on their graduation result using educational data mining" | Adekitan, A. I., and Salau, O. (2019).        | 102             | [39]     |
| 9    | "Analysis of student behavior in learning management systems through a Big Data framework"                                          | Cantabella, M., et.al. (2019).                | 97              | [137]    |
| 10   | "Teaching with analytics: Towards a situated model of instructional decision-making"                                                | Wise, A. F., and Jung, Y. (2019).             | 96              | [129]    |

Source: Authors' elaboration using Bibliometrix R package.

learning analytics, c) performance prediction, and d) analysis of academic behaviors, trajectories, and decision making explain the high resonance of the ten most cited papers.

The most cited article is "Digital academic entrepreneurship" by Rippa and Secundo (2019), with 189 citations indicating the great potential of digital technologies to foster entrepreneurship within academia, a topic of strategic interest for universities [141]. In second place, the systematic review "Utilising learning analytics to support study success in higher education: a systematic review" by Ifenthaler and Yau (2020) with 129 citations is a key reference, by rigorously synthesizing the existing evidence on the effectiveness of learning analytics interventions to improve student success [9]. In the third position, Rajabalee and Santally's (2021) study with 120 citations made an important contribution by revealing that satisfaction, motivation, and engagement are critical factors influencing performance in online education [40]. Fourth, Larrabee Sønderlund et al. (2019), with 120 citations, provide a much-needed global synthesis about what kind of analytics interventions have the greatest positive impact on student performance [132]. Francis and Babu (2019) with 118 citations propose a valuable hybrid data mining approach to predict academic outcomes,

laying a foundation for building such predictive models [90].

Mengash (2020), with 117 citations, presented a valuable data mining model to predict student performance and support university admission processes, demonstrating another application of predictive analytics [38]. The work of Abu Zohair (2019), with 105 citations, makes a methodological contribution by proposing effective predictive modeling techniques on small datasets, a frequent situation in educational contexts [142]. Adekitan and Salau (2019), with 102 citations, revealed important findings on how early performance in early college years has a strong impact on predicting outcomes at graduation, using educational mining techniques [39]. Cantabella et al. (2019), with 97 citations, presented a novel big data analysis framework to study in-depth learning behavior patterns, a promising approach [137]. Finally, Wise and Jung (2019) with 96 citations propose a situated model of teaching decision-making guided by learning analytics, laying conceptual foundations in this area [129].

In summary, we can state that the approach to BI and data analytics in the context of Higher Education Institutions (HEIs) has been the subject of extensive scrutiny and has aroused considerable interest in academic circles. The ten selected papers, highlighted by their frequent citations, accurately delineate the main areas of research and interest in the application and management of data to enhance the quality of higher education. These findings not only provide valuable guidance for practitioners and leaders seeking to enhance their skills and knowledge in the field of BI and data analytics but also serve as a source of inspiration for academics and researchers aspiring to contribute to this dynamic and evolving area.

### 3.5.1. Exploring BI and data analytics in higher education: sector and contexts (RQ6)

To identify the thematic areas linked to the study variable, this study applied a bibliometric method called "co-occurrence analysis" of words. The criteria applied to the VOSviewer software were type of analysis=Co-occurrence, Unit of analysis= author keywords, counting method = full counting, minimum number of occurrences of a keyword  $n = 5$ , number of keywords to be selected  $n = 79$ , and Method Normalization = Association; the result is shown in Fig. 10.

Table 8 presents a tabular representation of the data produced using VOSviewer. This table allows us to distinguish between several sectors or fields related to particular aspects of BI and data analytics in higher education institutions. The study organized these sectors into ten distinct groups based on related keywords.

Cluster 1 is highlighted by keywords such as classification [143], classification algorithms [142], decision trees and deep learning. This reflects its focus on the education and educational technology sectors, where data mining [38,90], and machine learning [91] techniques are applied. The aim is to develop predictive models that reveal patterns of student performance and attrition, playing a crucial role in the implementation of BIs systems in education to improve institutional decision-making.

Cluster 2 focuses on concepts such as big data [137,138], data science [144,145] and artificial intelligence [134,146], in the technology industry [48], where innovation and adoption of digital technologies is key. This highlights the use of advanced analytics to interpret voluminous educational data, which is crucial for creating BI systems in higher education that maximize the value of this massive amount of data.

Cluster 3 is characterized by concepts of educational innovation [147,148], blended learning [149,150], social network analysis [150, 151] and data visualization [152] reflecting an innovating education sector. It highlights the use of data from virtual learning platforms and academic social networks to complement quantitative analyses and enrich educational BI, offering a more complete view of student performance and experience.

Cluster 4 focuses on topics such as educational technology [153], e-learning [40,137] and instructional design [43,154], relating to the education and technology sectors, with emphasis on how educational



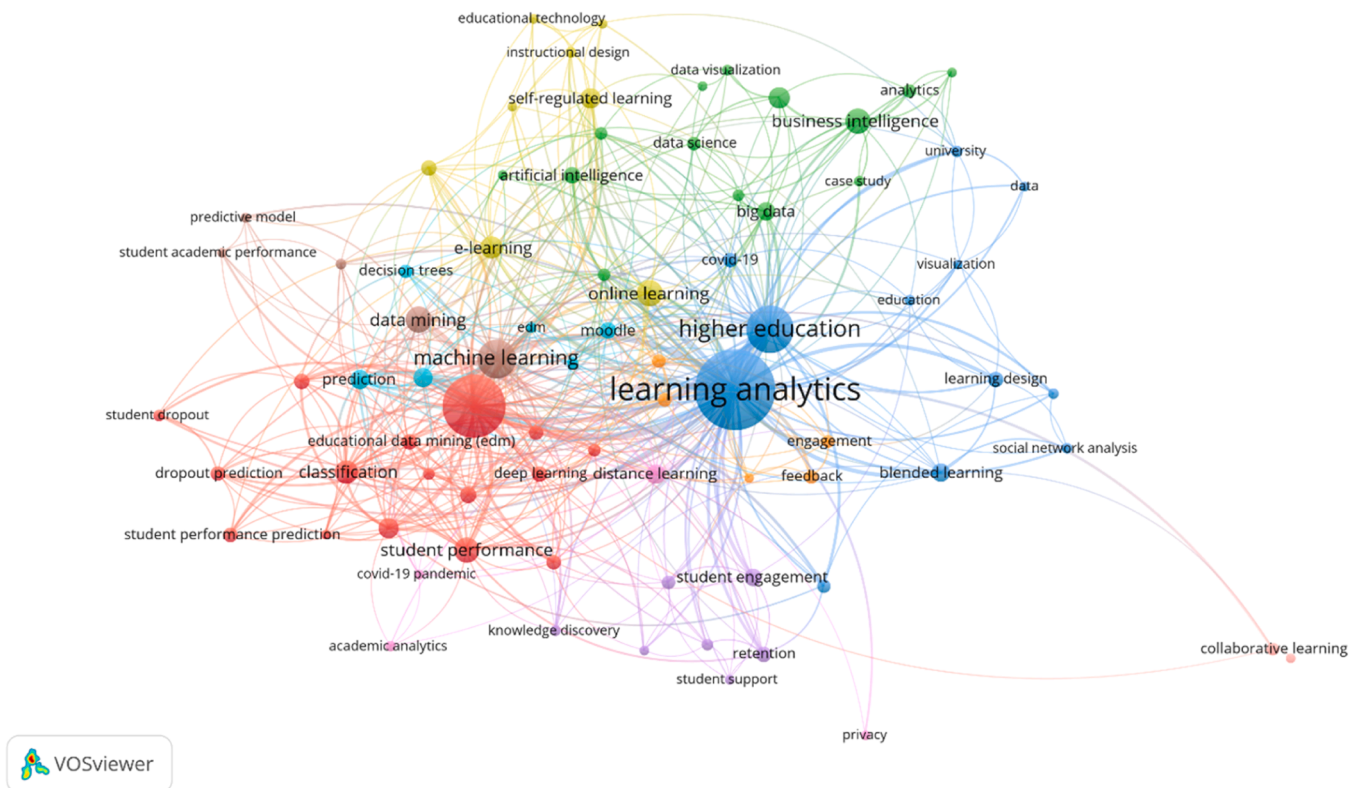


Fig. 10. Keywords related to BI and data analytics. Source: Authors' elaboration using VOSviewer.

platforms and methodologies can foster effective learning [155]. This emphasizes the importance of developing advanced educational analytics that use data from various digital platforms to enrich the online learning experience.

Cluster 5 is characterized by concepts such as distance education [119,120], student retention [156,27], and predictive learning analytics [118,119] with a focus on the virtual education sector. This highlights the importance of assessing factors that influence retention and effective online learning using predictive analytics to identify patterns of student persistence and engagement. These analytics are crucial to inform decisions regarding virtual program design, online curricula, and retention strategies.

Cluster 6 is distinguished by keywords such as academic performance [41,90], clustering [157,158], and prediction [38], focusing on the education and educational technology sectors to better understand student performance [90]. It highlights the importance of identifying patterns and factors that affect student success through data analysis, contributing to the development of BIs systems that optimize educational decision-making based on learning outcomes.

Cluster 7 focuses on improving assessments [2,159], student engagement [40,160], and engineering education [161,162], highlighting the integration of assessment and feedback data into educational BI systems [163]. He analyzed these metrics to gain insights into the effectiveness of teaching and assessment, identifying opportunities for improvement in formative processes, and evaluative methods in higher education and engineering.

Cluster 8 is closely linked to advanced concepts such as data mining [91], machine learning [138] and predictive modeling [164], specifically applied to academic performance in higher education [15]. This highlights the use of educational data mining and predictive models to uncover deep insights into learning processes, highlighting their importance in improving educational understanding and performance.

Cluster 9 stands out for its specific focus on crucial topics such as academic analytics [48,149], the impact of COVID-19 [149,165] and

privacy in distance education [166,167]. This cluster goes to the heart of distance education, thoroughly analyzing the impact of the pandemic and addressing privacy as an essential component.

Building on the identified clusters, specific implementations illustrated how BI systems have been successfully applied in higher education. For instance, Mengash (2020) [38] demonstrated the utilization of data mining techniques to optimize university admission processes, aligning with the goals outlined in Cluster 1. Similarly, Herodotou et al. (2020) [119] exemplified the application of predictive analytics to enhance student retention and engagement, directly supporting the themes of Cluster 5. These empirical examples provide a concrete perspective on how BI methodologies are operationalized to address institutional challenges and improve decision-making processes. Moreover, studies by Bordoloi et al. (2021) [149] and Zhang et al. (2020) [152] highlight the integration of blended learning environments and data visualization tools, underscoring their role in enhancing educational innovation and student outcomes, as discussed in Clusters 3 and 8. These cases not only validate the identified thematic clusters, but also demonstrate the actionable potential of BI in advancing personalized and data-driven strategies in higher education.

Finally, cluster 10 distinguishes itself by addressing advanced concepts, such as collaborative learning [151,168] and multimodal learning analytics [169,170]. It stands out for its focus on diversifying and enriching educational practices and the design of learning experiences, considering the integration of various types of data as the key to the development of advanced educational BI systems.

In summary, the analyses identified ten clusters in BIs and analytics in higher education, integrating areas such as educational technology, artificial intelligence, business intelligence, data science, educational innovation, online education, distance learning, educational data mining, engineering education, and machine learning. By focusing on specific topics, each cluster provides unique perspectives, advanced techniques, and complementary datasets that have the potential to enrich and expand the capabilities of BI and data analytics systems for

**Table 8**

Research streams in the related literature.

| Clúster | Keywords                                                                                                                                                                                                                                                                                                                                                                    | Sector and related context                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1       | Classification, classification algorithm, decision tree, deep learning, dropout prediction, educational data mining, educational data mining (edm), feature selection, learning management system, performance prediction, random forest, student dropout, student performance, student performance prediction, support vector machine, virtual learning environment (vle). | Education, Educational Technology, Artificial Intelligence applied to education. |
| 2       | Analytics, artificial intelligence, big data, business analytics, business intelligence, case study, data analysis, data analytics, data science, data visualization, decision making, educational big data, learning analytics dashboard, personalized learning.                                                                                                           | Business Intelligence (BI); Data Science in Education and Business.              |
| 3       | Blended learning, covid-19, data, dropout, education, educational innovation, higher education, learning analytics, learning design, social network analysis, university, visualization.                                                                                                                                                                                    | Education, Educational Innovation.                                               |
| 4       | Academic achievement, e-learning, educational technology, instructional design, learning management system, mobile learning, online learning, self-regulated learning.                                                                                                                                                                                                      | Education, Online Educational Technology.                                        |
| 5       | Distance education, knowledge discovery, predictive learning analytics, retention, student engagement, student retention, student support.                                                                                                                                                                                                                                  | Distance Education, Learning Analytics.                                          |
| 6       | Academic performance, clustering, decision trees, edm, moodle, prediction.                                                                                                                                                                                                                                                                                                  | Education, Educational Technology, Educational Data Mining (EDM).                |
| 7       | Assessment, engagement, engineering education, feedback, MOOC.                                                                                                                                                                                                                                                                                                              | Engineering Education, Online Education.                                         |
| 8       | Data mining, machine learning, predictive model, student academic performance, virtual learning environment.                                                                                                                                                                                                                                                                | Education, Machine learning.                                                     |
| 9       | Academic analytics, covid-19 pandemic, distance learning, privacy.                                                                                                                                                                                                                                                                                                          | Distance Education.                                                              |
| 10      | Collaborative learning, multimodal learning analytics.                                                                                                                                                                                                                                                                                                                      | Education, Educational Technologies.                                             |

Source: Authors' elaboration.

higher education. This multidimensional approach, which draws on expertise from diverse fields, promises significant boosts in the continuous improvement and adaptability of educational environments through the strategic application of technological tools and data analytics.

### 3.6. Analysis of methodological approaches to BI research and data analytics in HEIs: predominant methodologies, methods, and techniques (RQ7)

Using the bibliometrix R package and MS Excel, an analysis of the results of Keywords Plus, Keywords 'Authors, Keywords Title, and Keywords Abstract (Fig. 11) was performed to discern the terms related to the methodology and methods applied in the documents chosen for this research.

Examining the abstracts of 755 articles for these terms identified 35 categories of methodologies and methods used in BI exploration and

data analysis in institutions of higher education. In addition, the number of articles using each approach and the most prominent authors cited in relation to each methodology were determined (Table 9).

We now review and discuss the relevant data, focusing on the main observations and studies of the authors listed in Table 9. This analysis allowed us to delve deeper into the trends, conclusions, and methodological patterns present in the current research (Fig. 12), providing a richer and more detailed perspective on the topic at hand:

1. *Experimental Research*: This is the most used methodology with 109 papers, indicating a preference for approaches that allow controlling variables and determining causes and effects. Studies such as those by Buenaño-Fernández et al. (2019) [138] focused on the optimization of decision-making processes in BI, while Zhang et al. (2020) [114] explored advanced data analysis techniques to improve operational efficiency in higher education institutions.
2. *Quantitative Research*: The second most employed methodology, with 70 studies, this approach is exemplified in the work of Cerezo et al. (2020) [171] and Saqr and Alamro (2019), who used process mining and social network analysis, respectively, to improve the understanding and support of online learning processes.
3. *Random Sampling*: With 46 papers, Yağcı (2022) [91] and Zabriskie et al. (2019) [172] demonstrated how random sampling can be used effectively in education to ensure representativeness of samples and validity of statistical conclusions, especially in studies that seek to predict student achievement and tailor data-driven educational interventions.
4. *Survey*: Used in the 46 studies. Zhu and Wang (2020) explore the integration of intelligent teaching in English, combining learning analytics, surveys, ratings, and observations to evaluate collaborative mobile learning in STEM students [136]. Karkar et al. (2020) investigate the challenges of adopting e-learning at Kufa University, employing surveys and data mining, specifically clustering and decision trees [173].
5. *Machine learning*: In 2019, 46 studies employed this specialized method to predict student performance and analyze its impact on engineering education. Studies by Abu (2019) [142] and Adekitan and Salau [39], with 105 and 101 citations, respectively, illustrate the effective use of machine learning to improve educational outcomes and pedagogical decision making, highlighting its importance in academia.
6. *Predictive Analytics*: applied in 43 2020 studies, one by Hooshyar et al. (2020) [46] with 81 citations, and Gutierrez et al. (2020) [174] with 75 citations, both focused on predictive analytics in education. Hooshyar et al. investigated the prediction of academic performance, while Gutierrez et al. developed a learning analytics dashboard to improve academic decision-making.
7. *Data Mining*: Highlighted in 42 publications, emphasizing its use to predict students' academic success by analyzing large volumes of data. Mengash (2020) [38] and Francis and Babu (2019) [90] demonstrate the effectiveness of this technique in improving student selection and support processes in education. Both studies highlight the relevance of data mining in the optimization of educational and administrative management.
8. *Mixed research (Qualitative and Quantitative)*: used in studies by Rajabalee and Santally (2021) [40] and Jivet et al. (2020) [175] to investigate digital education with an emphasis on student satisfaction and engagement. With 119 and 59 citations, these papers highlight the importance of comprehensively understanding online learning experiences, showing how this methodology provides valuable insights for optimizing education in the digital age.
9. *Qualitative Research*: With a total of 34 studies applying this methodology, Wise and Jung (2019, 95 citations) [129]

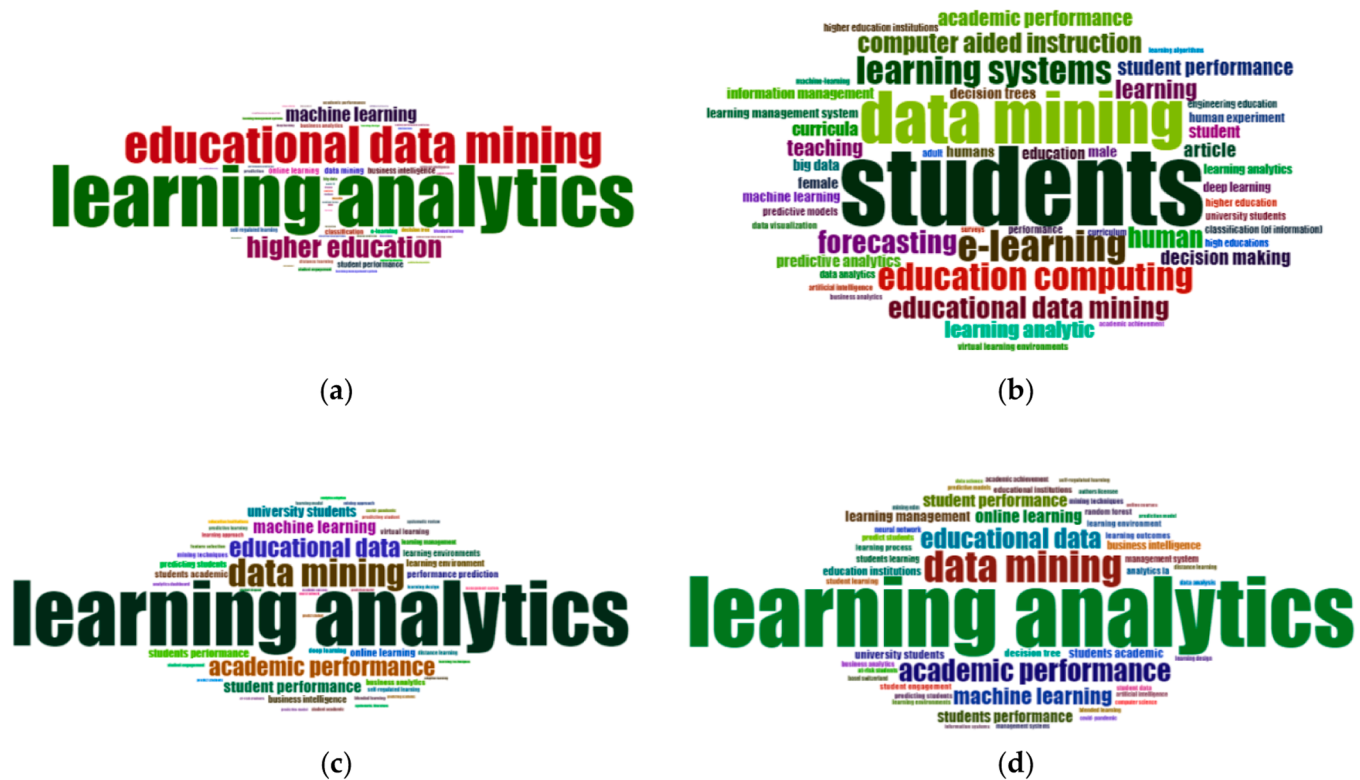


Fig. 11. Cloud of words: (a) Keyword's Author; (b) Keywords Plus; (c) Keywords Title (bigrams); (d) Keywords Abstract (bigrams).

- demonstrate how teaching can be improved by tailoring learning analytics to specific classroom and student needs. Tsai et al. (2020, 47 citations) [123] focused on empowering learners through personalized learning technologies and promoting an adaptive and learner-centered educational approach.
10. *Case Study*: This is the methodology used in 34 papers. Cantabella et al. (2019) [137] examine student behavior in the LMS of the Catholic University of Murcia using big data to evaluate access and tools. Dollinger et al. (2019) [176] study the co-creation of educational analytics platforms between designers and teachers in Australian universities.
  11. *Questionnaire*: applied in 25 investigations, one by Bordoloi and Ritimoni (2021, 71 citations) [149], and Sarra et al. (2019, 39 citations) [3], used questionnaires in studies of online education and academic failure risk prediction. Bordoloi explored the perception of online/blended learning during crises, while Sarra investigated the use of data mining to predict student achievement, highlighting the usefulness of questionnaires to obtain data on educational behaviors and attitudes.
  12. *Curriculum*: Methodology used in 24 documents. Williams (2019) [134] investigates the impact of emerging technologies on the university curriculum, suggesting the need for adaptations towards a more holistic approach to learning. Kollom et al. (2021) [125] analyzed academic expectations of learning analytics in Europe, seeing them as key to optimizing teaching and curricula.
  13. *Data analysis*: used in 24 studies, highlights its importance in academia. Combata et al. (2020) [37] develop a framework for data governance in universities, improving decisions and resource management. Zhang et al. (2022) [177] apply data mining to improve educational quality in university systems, identifying areas for improvement and supporting strategic decisions. Both reflect how data analysis benefits educational and administrative management in the university sector.
  14. *Descriptive Research*: applied to 21 studies to analyze educational phenomena. Ahn et al. (2019) [178] delved into the design of educational technologies from a usability and real context perspective, while Boticki et al. (2019) [45] investigated the behavior of e-book users, highlighting the need for personalization. Both studies highlight how this approach provides valuable insights for the design of more tailored and effective educational solutions.
  15. *Regression analysis*: applied in 20 studies. Adekitan and Noma-Osaghae (2019) predicted academic performance [15], and Hassan et al. (2019) to assess student attrition in virtual environments [143]. With 67 and 53 citations, respectively, both studies highlight the importance of this methodology in identifying critical factors that affect student success and retention, emphasizing its usefulness in improving educational practices.
  16. *Interviews*: applied in 17 studies Herodotou et al. use interviews to explore predictive analytics in education. Their first study (2019, 89 citations) [129] analyzed the impact of predictive analytics on large-scale learning, while the second (2021, 16 citations) [126] investigated the interaction of university teachers with these tools, highlighting the crucial role of human feedback. Both studies highlight the effectiveness of interviews in gaining in-depth insights into the implementation and perception of predictive analytics, thereby making valuable contributions to the field.
  17. *Literature Review*: used in 15 studies in 2019 to explore digital academic entrepreneurship and critically evaluate learning analytics. Rippa and Secundo (2019) [141] discuss how digital technologies transform education, whereas Selwyn (2019) [95] focuses on the ethical challenges and issues of learning analytics. Both highlight the importance of reviewing the literature to understand current and future trends in digital education.
  18. *Applied Research (Technology)*: This is a methodology used in 15 papers, highlighting the work of Guerra et al. (2020) [179] in the implementation of an analytics dashboard in Latin American universities to improve academic decision making, and that of



**Table 9**

Types of methodology/methods/techniques applied in this study.

| No. | Types of methodology/methods/techniques.    | No. Documents | Most cited authors of related studies.                                       |
|-----|---------------------------------------------|---------------|------------------------------------------------------------------------------|
| 1   | Experimental Research                       | 109           | Buenafío-Fernández et.al. (2019) [138]; Zhang et.al. (2020) [114].           |
| 2   | Quantitative Research                       | 70            | Cerezo et.al. (2020) [171]; Saqr and Alamro (2019) [150].                    |
| 3   | Random Sampling                             | 46            | Yağcı (2022) [91]; Zabriskie et. al. (2019) [172].                           |
| 4   | Survey                                      | 46            | Zhu and Wang (2020) [136]; Karkar et.al. (2020) [173].                       |
| 5   | Machine learning                            | 46            | Abu (2019) [142]; Adekitan and Salau (2019) [39].                            |
| 6   | Predictive Analytics                        | 43            | Hooshyar et.al. (2020) [46]; Gutiérrez et.al. (2020) [174].                  |
| 7   | Data mining                                 | 42            | Mengash (2020) [38]; Francis and Babu (2019) [90].                           |
| 8   | Mix Research (Qualitative and Quantitative) | 39            | Rajabalee and Santally (2021) [40]; Jivet et.al. (2020) [175].               |
| 9   | Qualitative Research                        | 34            | Wise and Jung (2019) [129]; Tsai et.al. (2020) [123].                        |
| 10  | Case Study                                  | 34            | Cantabella et al. (2019) [137]; Dollinger et.al. (2019) [176]                |
| 11  | Questionnaire                               | 25            | Bordoloi, Ritimoni (2021) [149]; Sarra et.al. (2019) [3].                    |
| 12  | Curriculum                                  | 24            | Williams (2019) [134]; Kollom et.al. (2021) [125].                           |
| 13  | Data analysis                               | 24            | Combata et.al. (2020) [37]; Zhang et.al. (2022) [177].                       |
| 14  | Descriptive research                        | 21            | Ahn et.al. (2019) [178]; Boticki et.al. (2019) [45].                         |
| 15  | Regression analysis                         | 20            | Adekitan and Noma-Osaghae (2019) [15]; Hassan et.al. (2019) [143].           |
| 16  | Interview                                   | 17            | Herodotou et. al. (2019) [129], 2021 [126]).                                 |
| 17  | Literature Review                           | 15            | Rippa and Secundo (2019) [141]; Selwyn (2019) [101]””.                       |
| 18  | Applied research (Technology)               | 15            | Guerra et.al. (2020) [179]; Kaliisa et.al. (2020) [180].                     |
| 19  | Theoretical framework                       | 14            | García-Peñalvo and Francisco (2021) [48]; Vargas et.al. (2019) [181].        |
| 20  | Longitudinal research                       | 11            | Williamson (2019) [133]; Salazar-Fernandez et.al. (2021) [182].              |
| 21  | Systematic review                           | 10            | Ifenthaler and Yau (2020) [9]; Larrabee et.al. (2019) [132].                 |
| 22  | Exploratory research                        | 9             | Shukor and Abdullah (2019) [43]; Murray et.al. (2020) [183].                 |
| 23  | Observation                                 | 7             | Han and Ellis (2020) [184]; Chiu et.al. (2019) [185].                        |
| 24  | Bibliometric                                | 6             | De Oliveira et.al. (2021) [186]; Oliva-Cordova et.al. (2021) [187].          |
| 25  | Online Survey                               | 4             | Li et.al. (2022) [188]; Burkardt et.al. (2019) [189].                        |
| 26  | Critical review                             | 4             | Jones (2019) [166]; West et.al. (2020) [190].                                |
| 27  | Empirical research                          | 4             | Jääskelä et.al. (2021) [191]; Lowes (2020) [192].                            |
| 28  | Cross-sectional Study                       | 3             | Amirhajlou et.al. (2019) [193]; Piri et.al. (2020) [33].                     |
| 29  | Conceptual framework                        | 3             | Cavanagh et al. (2020) [155]; Biswas et.al. (2020) [194].                    |
| 30  | Action research                             | 2             | Arnaboldi et.al. (2021) [195]; Suleman and Chigeza (2019) [44].              |
| 31  | Content analysis                            | 2             | Nkhoma et.al. (2020) [196]; Baldry et.al. (2023) [197].                      |
| 32  | Factor analysis                             | 2             | Faúndez and de la Fuente-Mella (2022) [198]; Nasa-Ngium et.al. (2023) [199]. |

**Table 9 (continued)**

| No. | Types of methodology/methods/techniques. | No. Documents | Most cited authors of related studies.                    |
|-----|------------------------------------------|---------------|-----------------------------------------------------------|
| 33  | Cohort Study                             | 2             | Kühbeck et.al. (2019) [200]; Qahmash et.al. (2023) [201]. |
| 34  | Group process                            | 1             | Dindar et.al. (2020) [202].                               |
| 35  | Quasi-experimental research              | 1             | Kay and Bostock (2023) [203].                             |

Source: Authors' elaboration using bibliometrix R package and MS-Excel.

Kaliisa et al. (2020) [180], which integrates analytics to support the design of blended learning, analyzing key student behaviors.

19. *Theoretical framework*: used in 14 studies, most notably García-Peñalvo and Francisco (2021) [48], which defines a framework for implementing e-learning in face-to-face universities, focused on ethical data management and academic analytics. Vargas et al. (2019) [181] present a framework to support competency-based education, facilitating competency assessment, learning analytics and identification of areas for improvement.
20. *Longitudinal Research*: was used in 11 studies, such as Williamson (2019) [133], who analyzed the impact of digital data on education, highlighted by 75 citations, and Salazar-Fernandez et al. (2021) [182] who studied school dropouts using a curricular approach and received nine citations. Both studies highlight the importance of this methodology in understanding the long-term effects of education.
21. *Systematic review*: used in 10 studies, including Ifenthaler and Yau (2020) [9], who reviewed 46 of 6220 articles to analyze the impact of learning analytics on academic success and student retention. Larrabee et.al. (2019) [132] evaluated 11 of 689 studies on learning analytics in educational interventions, highlighting their usefulness and emphasizing the need for more research to validate their effectiveness.
22. *Exploratory research*: Nine studies stood out for understanding educational phenomena. Shukor and Abdullah (2019) [43] analyze how learning analytics optimizes online courses, whereas Murray et al. (2020) [183] explored factors that impact teacher job satisfaction, revealing crucial insights for future research and practice.
23. *Observation*: Seven studies revealed the dynamics of student learning and interaction. Han and Ellis (2020) [184] combined observations and self-reports to analyze student engagement, whereas Chiu et al. (2019) [185] explored the impact of active peer engagement, demonstrating the usefulness of observation in understanding complex educational interactions.
24. *Bibliometric*: used in six studies, highlighting the work of De Oliveira et al. (2021) [186] on the prevention of student dropout through learning analytics and Oliva-Cordova et al. (2021) [187], which reviews the relationship between learning analytics and teaching skills at universities, both through systematic and bibliometric literature reviews.
25. *Online Survey*: used in four studies, with Li et al. (2022) [188] exploring disparities in the acceptance of educational data use, highlighting the importance of privacy and ethics in educational analytics. Burkardt et al. (2019) [189] examined the factors critical to the success of educational technologies, showing the value of understanding user perspectives. Both studies highlight the benefits of online surveys in investigating educational and technological issues.
26. *Critical review*: The methodology used in all four papers, including Jones (2019) [166], who examined privacy concerns in learning analytics and advocated for informed consent and more autonomy for students. West et al. (2020) [190] investigated the representation of student perspectives in learning analytics,

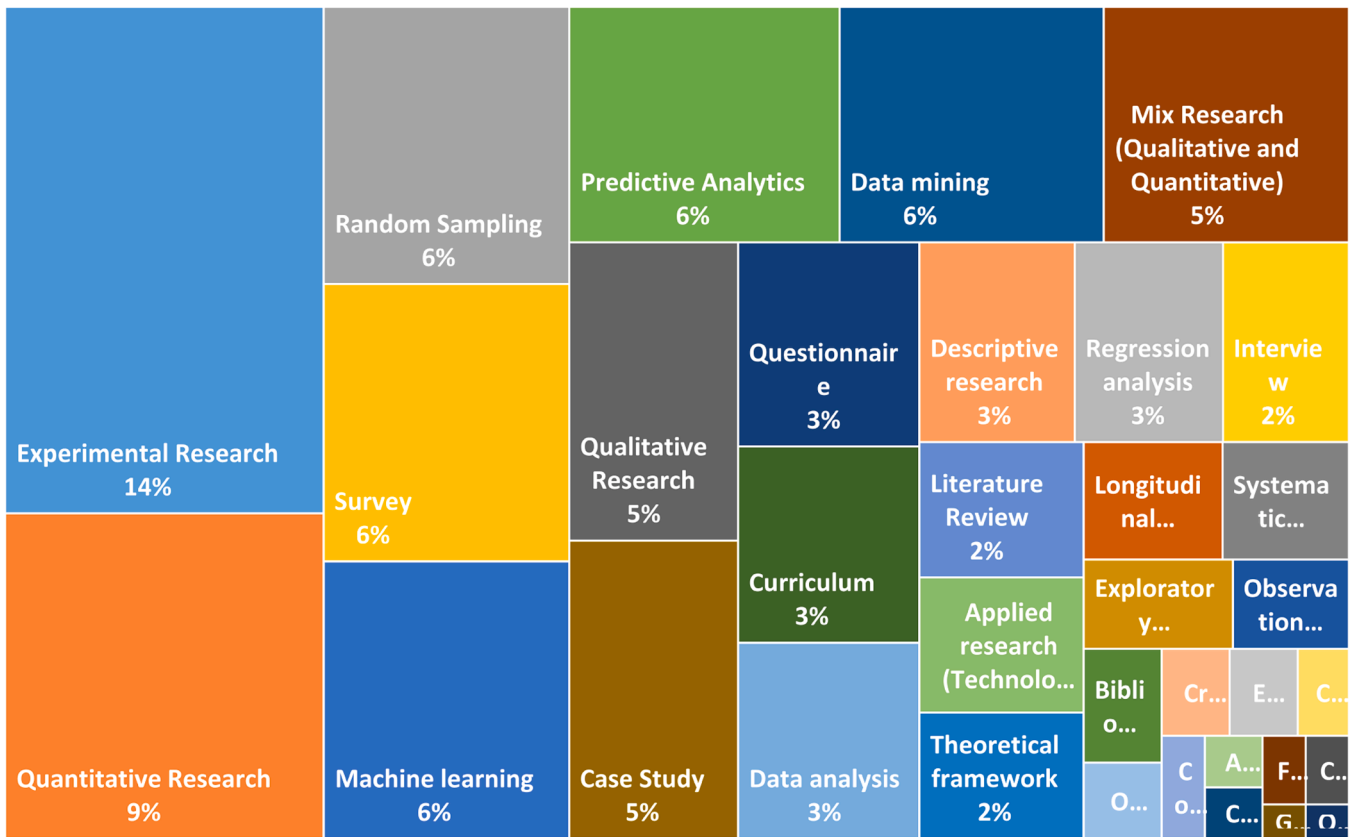


Fig. 12. Tree map of methodologies, methods and techniques distributed in the study.

- highlighting the importance of considering students as key participants in their development and ethical application.
27. *Empirical Research*: applied in four studies for the direct analysis of educational practices. Jäskelä et.al. (2021) [191] highlight how analytics can create participatory educational environments, and Lowes (2020) [192] highlights the value of personal tutoring and learning analytics for more personalized student support. Both reflect how empirical research drives innovation to enrich education.
  28. *Cross-sectional study*: Three studies used this methodology. Amirhajlou et al. (2019) [193] conduct a pilot study to predict resident performance using data mining and techniques such as neural networks and linear regression, on 841 residents in Iran. Piri et al. (2020) [33] applied this methodology to improve academic decision-making by collecting data from 30 academic managers and creating a digital dashboard with performance indicators.
  29. *The conceptual framework*: Used in all three studies. For example, Cavanagh et al. (2020) [155] develop a design framework for adaptive learning in higher education, identifying key features at the University of Central Florida. Biswas et.al. (2020) [194] proposed a hierarchical scheme for modeling apprentices in complex decision-making training, analyzing cognitive processes, and resolution strategies.
  30. *Action research*: Used in Two Papers. Arnaboldi et al. (2021) [195] applied it to design a performance measurement tool in a university, exploring BI dashboards. Suleman and Chigeza (2019) [44] used it to investigate factors affecting student engagement by analyzing data and experiences.
  31. *Content analysis*: used in two research studies. Nkhoma et al. (2020) [196] to analyze unstructured student data and Baldry et al. (2023) [197] to evaluate online video corpora in digital literacy, showing their impact on higher education.
  32. *Factor analysis*: used in two investigations to discover key variables in big data. Faúndez and de la Fuente-Mella (2022) [198] identify important factors in leadership skills, while Nasa-Ngium et al. (2023) [199] use factor analysis to analyze school dropout during the pandemic. Both highlight its effectiveness in educational and organizational research.
  33. *Cohort Study*: used in research by Kühbeck et al. (2019) and Qahmash et al. (2023) to analyze medical education. Kühbeck et al. [200] examine the relationship between online assessments and exam performance in pharmacology, while Qahmash et al. [201] study how admission criteria affect student performance in medicine, using statistics and data mining.
  34. *Group process*: This was used in the study by Dindar et al. (2020) [202] to analyze physiological synchronization in collaborative learning. This approach, with 35 citations, examined how group cohesion affects learning, emphasizing the importance of nonverbal and physiological factors in pedagogy and the design of effective educational environments.
  35. *Quasi-experimental research*: used in the study by Kay and Bostock (2023) to investigate the impact of technology and nudges on student achievement. Although not yet cited, it highlights the relevance of evaluating the effects of technological interventions in education and demonstrates how this methodological approach can inform the effectiveness of pedagogical innovations [203].
- BI and data analytics research in the management of higher education institutions has employed a wide variety of methodological approaches. Quantitative methods such as experimental research, random sampling, and surveys have been the most prominent, emphasizing the importance of collecting and analyzing representative and specific data from students and academic populations. In addition, researchers have used advanced techniques such as machine learning and predictive



analytics, demonstrating the growing interest in employing artificial intelligence and predictive analytics to enhance educational and decision-making processes. The COVID-19 pandemic further accelerated the adoption of data mining, as highlighted in 'Insights from the COVID-19 Pandemic,' demonstrating its value in addressing unforeseen challenges in educational management [22].

Nevertheless, the extensive utilization of quantitative methodologies may potentially overlook contextual and human-centered insights that qualitative approaches are uniquely positioned to provide. For instance, interviews, as observed in the studies by Herodotou et al. (2019, 2021), have been instrumental in elucidating the usability challenges of predictive analytics tools, offering crucial feedback from educators. Similarly, case studies, such as those conducted by Cantabella et al. (2019), provide in-depth insights into specific educational contexts, such as student behavior on learning platforms. Mixed-method approaches, as employed by Rajabalee and Santally (2021), combine the strengths of quantitative precision and qualitative depth, facilitating a comprehensive understanding of phenomena, such as student engagement in digital learning environments. These examples underscore the significance of diversifying methodologies to address complex educational challenges in a comprehensive manner.

In conclusion, the diversity of methodological approaches used in BI research and data analytics in higher education underscores the complexity of the field and the need for robust and flexible approaches to capture a wide range of variables relevant to the management of educational institutions.

Fig. 13, provided below, serves as an illustrative representation to enhance the clarity of the findings related to the nine research questions uncovered in this review.

#### 4. Discussion

This bibliometric study has revealed key trends in business

intelligence (BI) and data analytics research in academic management in higher education institutions over the past five years, addressing specific questions about the evolution of topics, scientific output, and methodological approaches. We assessed the main and relevant themes that guided this evolution: the most influential academic sources, the most prominent authors in terms of productivity and impact, the trends identified in the most cited articles, and the predominant methodological approaches in research. We also identified patterns in the academic literature, highlighting predominant trends and emerging unexplored sectors that promise to become future lines of research through applied scientific mapping [204]–[68].

The results highlight a clear and progressive evolution in topics related to Business Intelligence (BI) and data analysis in higher education. Over time, research has adapted to the emerging needs of educational institutions, particularly in areas such as learning analytics and educational data mining. The findings revealed a thematic shift from the use of predictive analytics and academic data mining in the early years (2019–2020) to the greater integration of artificial intelligence and machine learning-based approaches in recent years (2021–2023). Additionally, the research shows how some key areas, such as academic analytics, have lost relevance, whereas others, such as business analytics and student dropout prediction, have gained traction. However, despite this progress, there is a potential overreliance on predictive analytics and quantitative approaches. Although the recent integration of artificial intelligence and machine learning-based approaches has marked positive progress, this shift may overlook the importance of qualitative and contextual factors in education. A critical analysis of how these technologies affect various institutions, especially those with limited technological capabilities, in the long term remains crucial.

Prestigious journals such as the 'British Journal of Educational Technology' and the 'Journal of Learning Analytics,' ranked in the Q1 quartile, have been the most influential, with important contributions from authors such as Christothea Herodotou [7,96,100,118,119] and

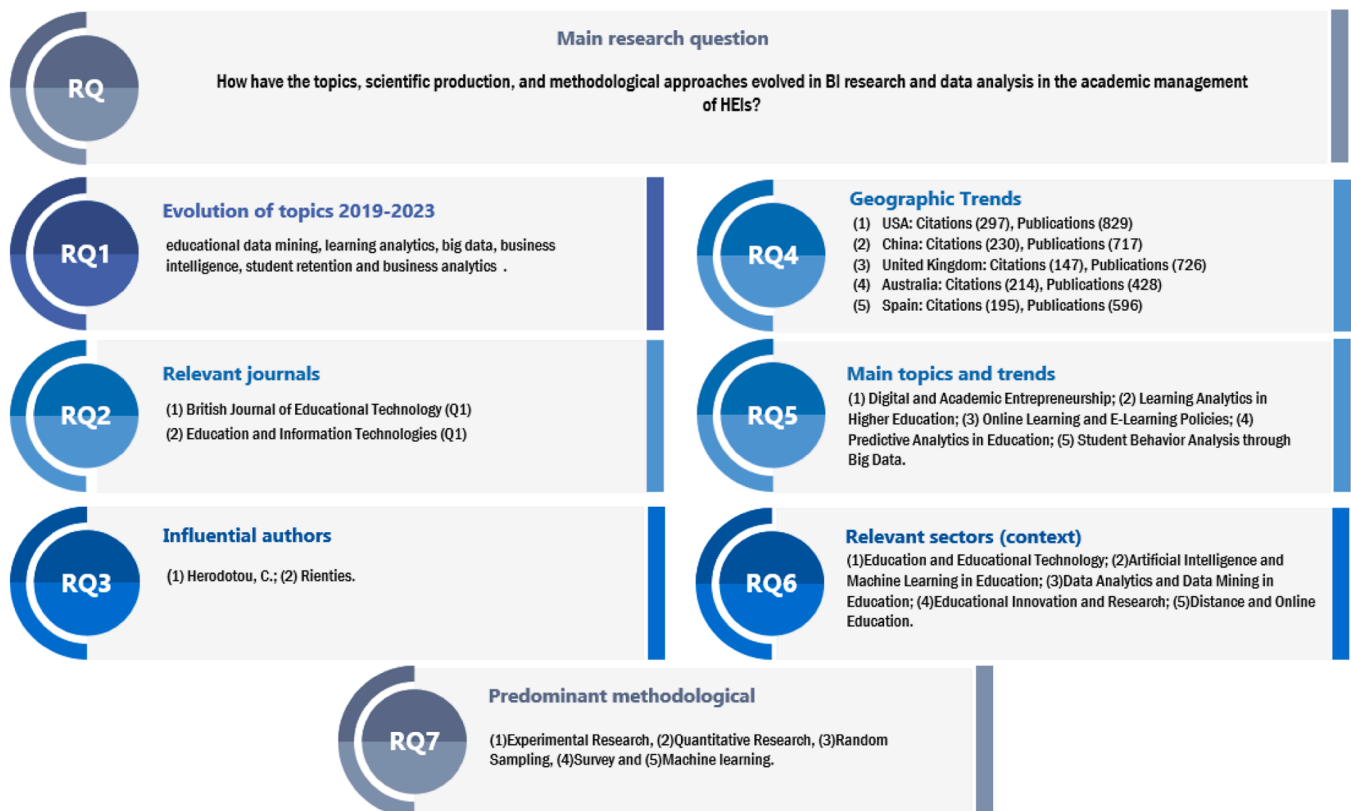


Fig. 13. Synthesis of results.

Bart Rienties [109,111,120]. These publications have consolidated topics in academic literature, reflecting a high impact and relevance in the field. However, the lack of comparative studies across different publication platforms and geographic regions remains challenging. This omission could limit the scope of the findings, especially in contexts in which technological platforms and access to resources are unequal. Institutions such as The Open University in the UK and Monash University in Australia have been prominent in the development of these fields, underscoring the global relevance of this study.

The bibliometric analysis reveals significant geographic trends in BI and data analytics research within higher education institutions, highlighting disparities in academic output and influence. Countries such as the United States, the United Kingdom, and China demonstrate leadership in publication volume, reflecting robust research infrastructures. However, a closer examination uncovers disparities in impact, as measured by citation counts. For instance, while the United States has the highest number of publications (829), its relative citation impact (297) suggests a potential dilution of influence due to the extensive breadth of research topics. Europe emerges as the most prolific continent, with contributions from 32 countries, indicating a concentrated focus on advancing BI and data analytics. Conversely, regions such as Africa, South America, and the Middle East exhibit limited representation, with fewer publications and lower citation impacts. This underscores a need for fostering inclusivity and promoting research in underrepresented regions.

These findings emphasize the critical need for geographically inclusive research agendas that address diverse educational challenges. Integrating perspectives from non-dominant regions could offer novel insights into how BI and data analytics can adapt to varying cultural, technological, and institutional contexts, thereby enriching the field's global applicability. Future research should prioritize collaboration across regions to bridge these gaps and ensure equitable contributions to the growing body of knowledge in BI and data analytics.

The most cited articles in Business Intelligence (BI) and data analytics in higher education institutions (HEIs) stand out in key areas such as digital academic entrepreneurship, learning analytics, and student performance prediction. Rippa and Secundo (2019) [141], with 189 citations, explore how digital technologies drive academic entrepreneurship, while Ifenthaler and Yau (2020) [9], with 129 citations, review the effectiveness of learning analytics interventions on student success. Rajabalee and Santally (2021) [40], with 120 citations, stress the importance of satisfaction and engagement in online learning environments. Other studies, such as those by Francis and Babu (2019) [90], Mengash (2020) [38], and Abu Zohair (2019) [142], with >100 citations each, develop predictive models based on data mining to support admission decisions and improve student achievement. Adekitan and Salau (2019) [39], Cantabella et al. (2019) [137] and Wise and Jung [129] (2019) explored academic behavior patterns to guide more personalized pedagogical decisions. While these studies have made important advances, a recurring criticism is the lack of integration of qualitative approaches that complement predominant quantitative models.

Although data mining and predictive models have proven to be powerful tools, the absence of methodologies that explore the more subjective aspects of academic behavior limits the depth of analysis. Future studies should not only consider combining quantitative and qualitative approaches, but also investigate the ethical and social impacts of the technologies applied, such as possible bias in performance prediction systems. Additionally, addressing the gaps in comparative studies and interdisciplinary methodologies is essential for advancing the field. Sun (2023) highlighted how cultural and technological disparities across regions can influence the effectiveness of BI applications, emphasizing the importance of context-sensitive solutions [205]. Similarly, Speziali et al. (2023) suggest that a broader analysis of the technological landscape could support the development of scalable and adaptable BI tools [206]. Du et al. (2022) further propose combining

technological approaches with educational and sociological theories to create more inclusive and effective methodologies [54]. These strategies can bridge the current gaps and provide actionable insights for diverse educational environments.

The integration of Business Intelligence (BI) and data analytics in higher education demonstrates significant potential to address institutional challenges and enhance innovation. The analysis identified ten thematic clusters, revealing critical areas such as predictive modeling, educational technologies, and distance education.

1. *Predictive Modelling and Decision-Making*: Clusters such as Cluster 1 highlight the role of predictive models, including classification algorithms and dropout prediction systems, in implementing early warning mechanisms to reduce student attrition [143,150]. Ensemble learning methods, as elucidated in Data-driven Interpretable Ensemble Learning Methods, augment the interpretability of predictive models, thereby enhancing their applicability to educational decision-making processes [207]. Similarly, Cluster 5 underscores the application of predictive analytics to enhance student retention and engagement in distance education [110,129]. For instance, the study 'AI in Higher Education' demonstrates how predictive analytics, supported by advanced algorithms such as PSO-WOA, enhances personalized learning and institutional performance. These methodologies illustrate the potential for integrating advanced AI-driven tools in higher education [208]. These tools improve institutional decision-making and align strategies with student-success objectives.
2. *Technological Innovation in Education*: Clusters such as Cluster 3 and Cluster 4 emphasize innovation in online learning and instructional design [157,162]. For instance, data from virtual learning environments and academic social networks provide actionable insights for personalizing learning experiences and improving engagement [158,160].
3. *Ethical and Emerging Concerns*: Cluster 9 addresses the growing need to balance innovation with ethical data usage, particularly during the COVID-19 pandemic [128,174]. This cluster highlights the importance of privacy frameworks for ensuring the responsibility of BI applications.

These findings underscore the transformative role of BI in academic management, from improving retention strategies to fostering personalized learning. Future research should focus on interdisciplinary collaborations and ethical frameworks to maximize the benefits of BI and data analytics, while addressing privacy concerns and technological limitations [60,144]. Moreover, future studies should investigate the potential application of advanced Artificial Intelligence (AI) techniques, as discussed in 'AI in Higher Education,' to implement adaptive learning strategies, thereby enabling more individualized and efficacious educational experiences [208].

The results of research on the predominant methodological approaches in Business Intelligence (BI) and data analytics in higher education institutions highlight a strong inclination towards quantitative methods, including experimental research, data mining, and machine learning, which optimize decision-making processes and predict academic performance. These methodologies align with thematic areas focused on improving educational processes and institutional management. Investigations such as those conducted by Buenaño-Fernández et al. (2019) [138] and Zhang et al. (2020) [114] illustrate the efficacy of experimental research in augmenting operational efficiency and decision-making in HEIs.

Nevertheless, the increasing use of mixed methodologies, combining quantitative and qualitative approaches, provides a more complete understanding of educational phenomena, particularly in learning analytics. For instance, Rajabalee and Santally (2021) [40] and Jivet et al. (2020) [175] elucidate how mixed methods reveal more profound insights into student satisfaction and engagement. Furthermore,

Herodotou et al. (2019) [129] demonstrated through interviews with educators how usability concerns in predictive analytics tools can impede adoption, emphasizing the significance of user-centered design. Case studies, such as those conducted by Dollinger et al. (2019) [176], further illustrate how participatory strategies facilitate addressing resistance and aligning BI tools with institutional requirements.

Although quantitative methods have demonstrated efficacy, this trend may present potential risks if not counterbalanced with enhanced integration of qualitative approaches that capture the more nuanced complexities of learning. Furthermore, it is imperative to adapt analytical models to diverse educational and cultural contexts across regions, as uniform approaches may prove ineffective in heterogeneous or resource-limited settings. In resource-constrained environments, challenges such as inadequate infrastructure, insufficient training, and resistance to adoption further complicate the implementation of Business Intelligence (BI) tools. The development of scalable and culturally appropriate frameworks is essential to surmount these barriers and optimize the global impact of BI systems. In fact, recent studies, such as those on Enhancing COVID-19 Vaccination Strategies, highlight the applicability of metaheuristics in optimizing complex systems, suggesting the potential for similar approaches in educational resource management [209].

Building on these observations, future research could expand its scope to include underexplored areas, such as the role of Business Intelligence (BI) in fostering equity and inclusion in education. While existing studies have primarily emphasized the predictive capabilities of BI in identifying at-risk students and enhancing retention strategies (Mengash, 2020 [38]; Guerra et al., 2020 [179]), its potential to address systemic inequities remains largely unexplored. Interdisciplinary methodologies from fields such as sociology, psychology, and economics could provide critical insights into the broader impacts of BI in education. For instance, sociology can analyze cultural and structural factors influencing the adoption of BI tools in diverse contexts, while psychology contributes to understanding student behavior, motivation, and engagement patterns derived from analytics. Similarly, economics offers frameworks for evaluating the cost-effectiveness and scalability of BI implementations in resource-constrained settings. By integrating these disciplines, BI research can develop tools that are not only effective but also equitable, addressing challenges in heterogeneous educational environments.

Additionally, emerging technologies such as adaptive learning systems provide a promising avenue for advancing personalized education. Powered by advanced algorithms and data analytics, these systems dynamically adapt instructional content and methods to individual learning needs (Zhang et al., 2023 [93]; Surahman et al., 2019 [210]). By offering scalable solutions, adaptive learning technologies address diverse educational challenges and contribute to a more equitable and context-sensitive approach to higher education. Frameworks that integrate learning analytics with competency-based systems can further enhance personalized interventions, creating pathways to improve student outcomes and institutional strategies (Alfonso et al., 2019 [211]). These innovations, combined with interdisciplinary methodologies, position BI as a fundamental component of inclusive and transformative educational practices.

#### 4.1. Implications

This study provides a broad overview of the evolution of business intelligence (BI) and data analytics issues in the academic management of higher education institutions (HEIs) over the past five years. However, one limitation is the lack of exploration of how these trends interact with established educational theories, which prevents their linkage with existing theoretical models in education and management.

On a practical level, BI's potential to improve decision-making is evident, but its implementation varies among HEIs, especially in countries with limited resources. Addressing this disparity requires

actionable strategies for practitioners and policymakers, such as capacity-building programs, investment in scalable technologies, and institutional frameworks, to guide BI adoption across diverse educational contexts. Practical case studies demonstrating effective BI applications are essential for generalizing the findings to a wider variety of institutions, including large universities, community colleges, and online learning platforms.

Methodologically, this study employed robust bibliometric tools; however, its exclusive reliance on the Scopus database may have excluded pertinent research from other platforms. Furthermore, the emphasis on quantitative methods, such as data mining and machine learning, results in a deficiency in the qualitative comprehension of user experiences. The integration of qualitative approaches and implementation of comparative studies across regions and cultures would yield a more comprehensive perspective on the impact of BI technologies and their adaptation to diverse educational and cultural contexts.

From an ethical perspective, BI adoption raises critical concerns regarding privacy, data bias, and equitable access. Mutimukwe et al. (2022) highlight the dimensions of students' privacy concerns in learning analytics systems, emphasizing the importance of robust frameworks to ensure transparency and safeguard sensitive information [112]. Institutions must adopt measures such as data anonymization, informed consent protocols, and compliance with regulatory frameworks like the General Data Protection Regulation (GDPR) to protect sensitive data and maintain stakeholder trust. Without these safeguards, the misuse of data could undermine confidence in BI systems and hinder their adoption.

Similarly, Brown and Klein (2020) observe that intelligent learning systems, which rely heavily on historical data, are prone to amplifying existing inequalities due to embedded biases in training datasets [167]. Addressing these issues requires regular algorithmic audits, the application of fairness-aware machine learning techniques, and diversifying datasets to reduce systemic biases. Combata Niño et al. (2020) further emphasize the role of governance frameworks in ensuring that BI tools are deployed with transparency, accountability, and stakeholder engagement [37]. These frameworks are essential for balancing innovation with ethical principles and promoting equitable access to BI technologies across institutions with varying resource levels.

Emerging areas, such as learning analytics and educational data mining, demonstrate potential; however, their long-term impacts require further investigation. Research should examine their applications in institutions of varying sizes, geographic regions, and teaching modalities. The integration of quantitative and qualitative methodologies will facilitate a more comprehensive understanding of BI implementation, while addressing ethical concerns and fostering equitable solutions. By focusing on privacy, bias, and governance, future research can bridge the gap between technological innovation and ethical responsibility, ensuring impactful and sustainable BI applications in higher education.

The findings of this study have significant implications for policymakers, educators, and technologists, offering actionable insights to enhance the practical application of Business Intelligence (BI) and data analytics in education. For policymakers, geographical analysis highlights disparities in the adoption and development of BI tools across regions. Addressing these disparities necessitates the creation of equitable frameworks, investment in infrastructure, and capacity-building programs to bridge the digital divide, particularly in underrepresented regions. To further enhance inclusivity, future research should prioritize the exploration of diverse cultural, institutional, and technological contexts. Specifically, resource-limited regions often encounter unique challenges such as insufficient infrastructure, limited access to digital tools, and a lack of technical expertise. Tailored BI solutions that account for these constraints can facilitate bridging gaps and foster equitable access. Furthermore, understanding the cultural factors influencing the acceptance and implementation of BI tools is critical for designing applications that are both effective and context-sensitive.

Comparative studies and interdisciplinary approaches that incorporate insights from sociology, economics, and psychology will be instrumental in developing frameworks adaptable to various educational environments globally.

Although this study establishes a robust foundation for understanding the evolution of BI and data analytics in HEIs, its limitations underscore the need for ongoing, inclusive, and context-aware research. Such efforts will not only ensure ethical and effective data management in academic institutions but also address the challenges and opportunities unique to diverse educational contexts.

#### 4.2. Limitations

As with any research endeavor, this study has certain limitations that subsequent investigations may address more comprehensively. For instance, the restriction of exploration to a single research database such as Scopus may impose constraints, as relevant studies published on other platforms such as Web of Science, IEEE, or regional repositories may be excluded. This limitation could potentially narrow the scope of the findings, particularly in underrepresented regions or emerging research communities. Future studies should broaden the range of data sources to provide a more inclusive perspective on BI and data analytics in education.

Furthermore, the use of broad search terms such as "business intelligence," "business analytics," "academic analytics," "learning analytics," "educational data mining," and "educational big data" may inadvertently exclude valuable studies that employ alternative terminologies or focus on niche areas. Although this approach captures general trends, it may overlook more specific or emerging themes in the field.

The reliance on bibliometric analysis also emphasizes quantitative metrics, such as citation counts and publication volume, which may not fully capture the qualitative dimensions of BI applications. This focus could lead to an overrepresentation of well-established topics and regions while underestimating practical and contextual nuances in diverse educational settings. For example, thematic clustering, which is effective in identifying key trends, is subject to algorithmic biases that may skew the results toward dominant research areas or authors.

Lastly, despite the comprehensive analysis of 755 peer-reviewed articles, this study does not include conference papers, which often reflect cutting-edge developments in nascent fields, such as BI in education. Incorporating these sources into future research can provide richer insights. To mitigate these limitations, future studies should consider complementing bibliometric methods with qualitative approaches such as expert interviews or case studies to explore the contextual and practical impacts of BI across diverse educational contexts.

#### 5. Conclusions

This bibliometric analysis uncovered key findings on business intelligence (BI) and data analytics in higher education institutions, demonstrating the evolution and adaptation of these fields across various academic contexts. It highlights the significant influence of academic publications and prominent researchers in the consolidation and expansion of these topics. The research also revealed geographic patterns in the study of these fields, pinpointing centers of interest and development in various regions worldwide. By examining the most highly cited articles, this study uncovered emerging themes and trends that indicate areas of increased interest and potential for future research on BI and data analytics in higher education. In addition, it explores practical applications across academic domains to clarify their scope and relevance. Notably, the study emphasizes the diversity of methodological approaches adopted, offering a comprehensive overview of the research to date and displaying a variety of approaches and depths of interest in these fields, significantly contributing to the understanding and expansion of knowledge in higher education.

This research has addressed the main question of how topics, scientific output, and methodological approaches in BI research and data analysis in academic management in HEIs have evolved. To answer this question, this study presents the following specific questions.

RQ1: What are the main and relevant themes that have guided the evolution of the approach to BI and data analytics in academic management of HEIs over the past five years? The research identified that the main and relevant themes in BI and data analytics in HEIs over the past five years include learning analytics, educational data mining, and business intelligence. During 2019–2020, the focus was on predicting academic performance using data mining and learning analytics. In 2021–2022, the topics diversified, incorporating feedback, student retention, and the impact of the pandemic on online learning. By 2023, researchers emphasized nonintrusive methods and deep learning models for predicting student performance.

RQ2: What are the most relevant academic sources in the field of BI and data analytics, as measured by their impact and importance according to the number of publications, h-index, and total citations? The most influential academic sources in the field of BI and data analytics are prestigious journals, such as the 'British Journal of Educational Technology' and the 'Journal of Learning Analytics,' ranked in the Q1 quartile. These publications have been instrumental in consolidating topics in academic literature, reflecting their high impact and relevance in the field.

RQ3: Who are the most influential authors in BI and data analytics, and how do their contributions stand out in terms of productivity and impact? Authors such as Christothea Herodotou and Bart Rienties have highlighted their significant contributions in terms of productivity and impact. Their work set benchmarks in BI research and data analytics, showing great influence on the development of these fields.

RQ4: How is geographically distributed BI and data analytics research and what trends emerge when comparing different countries and continents? The geographic analysis reveals a pronounced concentration of research outputs from the United States, China, and Europe, indicative of well-established academic ecosystems and collaborative networks. However, regions such as South America, Africa, and the Middle East demonstrate comparatively limited contributions, highlighting the necessity to incorporate more diverse studies from underrepresented contexts. Promoting research initiatives in these regions would enhance inclusivity and yield more comprehensive, varied perspectives on BI and data analytics in higher education. The expansion of global collaboration and the promotion of equitable access to research opportunities are essential steps toward a more holistic understanding of this field.

RQ5: What are the main themes and trends identified in BI and data analytics research based on an analysis of the most cited papers worldwide? The main identified trends reveal synergy with sectors such as education and educational technology. Methodological approaches such as experimental research, quantitative research, and machine learning predominate. Additionally, emerging areas such as mobile learning, business analytics, and academic and student performance assessments are emerging as guides for future research.

RQ6: How does the education sector interact with BI and data analytics in various contexts? The analysis identified ten thematic clusters that reflect critical areas of BI and data analytics in higher education, emphasizing predictive modelling, educational innovation, and online learning. Predictive tools such as dropout prediction models enable institutions to implement early warning systems, thereby reducing student attrition. Similarly, in distance education, analytics supports tailored strategies to improve retention and engagement, while machine-learning models optimize resource allocation and institutional performance.

These findings underscore the transformative potential of BI and analytics in enhancing decision-making processes and improving educational outcomes in higher education. By integrating advances in machine learning, data science, and educational technology, institutions can address critical challenges and foster innovation in academic



management and student success strategies.

RQ7: What are the predominant methodological approaches in BI and data analytics research, and how do they relate to the topic areas addressed? The studies reviewed in this analysis predominantly employed quantitative approaches, such as experimental research and the use of data mining and machine learning techniques. These approaches are valuable for identifying patterns and trends, but there is a gap in the qualitative understanding of the experiences and perceptions of BI and data analytics users of HEIs. Integrating qualitative approaches, in future research could provide a deeper and more holistic view of how these technologies impact educational stakeholders.

In conclusion, this study provides a solid foundation for understanding the evolution of BI and data analytics in HEIs, highlighting both the ongoing digital transformation and the challenges and opportunities that require attention to maximize the positive impact of these technologies. However, there are limitations, such as the lack of qualitative and comparative studies and the need to include greater diversity in academic sources and perspectives from different educational contexts. Future research should focus on overcoming these barriers, exploring new applications to improve the efficiency and quality of higher education, and ensuring ethical and effective data management in academia.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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